



Disorders of the Adrenal Gland

Note: This reference document contains the content of the Canvas page for “Disorders of the Adrenal Gland” with minor modifications for presentation in a document format. Refer to this session’s Canvas page for the full interactive experience or if any hyperlinks do not function properly.

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Overview

This session is built to reinforce the material you had read about in a prior teaching/learning platform. You could easily relate to the case-based discussion using prior knowledge.

During this session you will be working through a few cases which will help you integrate Pathology, Pharmacology, and Clinical Medicine.

Unless otherwise noted, all content and images for this lecture are adapted from ScholarRx and used with permission.

Learning Objectives

By the end of this session, you will be able to meet the following learning objectives:

1. Understand the Pathogenesis and Clinical Features of Cushing's Syndrome.
2. Describe the pathophysiology of endogenous Cushing's syndrome, including adrenal adenomas, pituitary adenomas, and ectopic ACTH production.
3. Describe the Pathology of Adrenal Adenomas and Carcinomas.
4. Understand the Pathogenesis and Clinical Features of Conn's Syndrome.
5. Describe the etiology and pathogenesis of primary adrenal insufficiency, including autoimmune destruction, infection, and genetic disorders.
6. Define Conn's syndrome (primary hyperaldosteronism) and describe the underlying pathophysiology, including aldosterone-producing adrenal adenomas and bilateral adrenal hyperplasia.
7. Define pheochromocytoma and describe the pathophysiology of catecholamine hypersecretion.

Prerequisites

Before progressing with this session, please review the following materials. *Refer to the Canvas page for the link to this course:*

Semester 4, ENRE

- Fusion Session | TBL: Structure and Function of the Adrenal Gland

Preparation Content

For this session, content has been sectioned into separate pages. *Refer to the Canvas page to visit these pages to review content for each topic. Return to the landing page to take the quiz after reviewing all content pages.*



- Fusion Session | Workshop: Disorders of the Adrenal Gland **Part 1 of 6: Adrenal Insufficiency**
- Fusion Session | Workshop: Disorders of the Adrenal Gland **Part 2 of 6: Hyperaldosteronism**
- Fusion Session | Workshop: Disorders of the Adrenal Gland **Part 3 of 6: Hypoaldosteronism**
- Fusion Session | Workshop: Disorders of the Adrenal Gland **Part 4 of 6: Congenital Adrenal Hyperplasia**
- Fusion Session | Workshop: Disorders of the Adrenal Gland **Part 5 of 6: Pheochromocytoma**
- Fusion Session | Workshop: Disorders of the Adrenal Gland **Part 6 of 6: Neuroblastoma**

Part 1 of 6: Adrenal Insufficiency

Learning Objectives

This page covers the following learning objectives:

1. Understand the pathogenesis and clinical features of Cushing's Syndrome.
2. Describe the pathophysiology of endogenous Cushing's syndrome, including adrenal adenomas, pituitary adenomas, and ectopic ACTH production.
5. Describe the etiology and pathogenesis of primary adrenal insufficiency, including autoimmune destruction, infection, and genetic disorders.

What Is Adrenal Insufficiency?

Case Connection

After Thanksgiving dinner, your aunt HK excuses herself to go lie down for a nap. Uncle RK says, "She hasn't been herself lately. She comes home from work and goes to sleep. She used to be busy—walks with neighbors, yoga class, book club—but she is always too tired now." You note that she looks thinner, and she confirms that she doesn't eat much because her stomach hurts. Later you talk to HK, and she admits that she has had body aches and low energy for the past year or so; lately, she's been extremely exhausted. She also has odd darkening in her mouth and gums. She thought it was because of menopause because her periods were less frequent, but says, "I'm only 44; isn't that too young?" She wonders if she should get tests done and if there is a way to treat her low energy.

What would be the best lab evaluation for HK? Consider your answer as you read, and we'll revisit HK at the end of the brick.

Adrenal insufficiency (AI) is a condition in which a person's adrenal glands don't produce enough cortisol and other key steroid hormones, leading to a variety of systemic effects. Causes can be autoimmune, infectious, or iatrogenic. The condition can be lethal, because a key role of cortisol is to maintain blood pressure and serum glucose levels, both needed for life.

AI is particularly common in patients who are acutely ill. It's estimated that 20% to 50% of patients in intensive care units (ICUs) have it to some degree, as do 75% of those in septic shock (ie, systemic infection with low blood pressure).

Classification

AI can be classified as follows:

- **Primary AI** occurs when the problem is an impairment of the adrenal glands themselves. It is often referred to as Addison disease in its chronic form.
- **Secondary AI** is actually much more common. It results when a disordered pituitary gland cannot produce and secrete adrenocorticotrophic hormone (ACTH). This is most often caused by patients taking exogenous glucocorticoid drugs that suppress ACTH secretion and normal adrenal function.
- **Acute AI** (or adrenal crisis or Addisonian crisis) can occur in either primary or secondary AI. Acute AI is a medical emergency in which there is an acute stressor like infection, and the body is unable to produce sufficient cortisol to maintain homeostasis.

Adrenal Hormones

Let's begin with cortisol, which is important in maintaining blood pressure, keeping the glucose level high enough, and regulating metabolism. It is one of the fight-or-flight hormones, like epinephrine (adrenaline). Cortisol is released in response to the anterior pituitary ACTH, part of the hypothalamic-pituitary-adrenal axis (HPA). In the hypothalamus of the brain, corticotropin-releasing hormone (CRH) is released and stimulates the anterior pituitary gland to release ACTH (**Figure 1**).

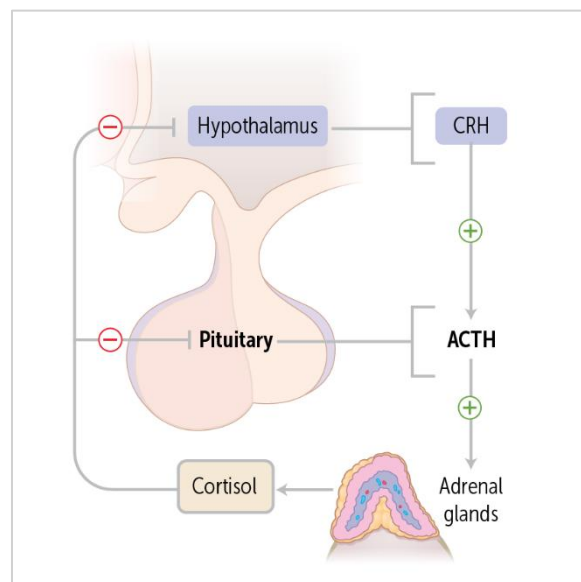


Figure 1. Image Credit: ScholarRx, used with permission.

This response occurs in reaction to a variety of body stressors, including hypotension, fear, and trauma. The cortisol produced in the adrenal cortex feeds back to suppress production of CRH and ACTH.

Clinical Correlation

Note that prescribed corticosteroids (which act like cortisol) will also suppress CRH and ACTH release, so can inhibit normal adrenal gland function. That will be an important factor in some cases of AI.

ACTH also has some effect on increasing the adrenal synthesis of mineralocorticoids, such as aldosterone; these hormones cause the kidney to retain sodium and help maintain blood pressure by increasing intravascular volume. They are mostly regulated by the renin-angiotensin-aldosterone system. In addition, ACTH increases the sex androgens, such as androstenedione, but these are mostly regulated by FSH and LH from the anterior pituitary gland.

In this brick, we primarily focus on the negative effects of low cortisol secretion. But just remember that AI can also reduce secretion of the other hormones through dysfunction of the adrenals themselves (primary) or by inadequate ACTH (secondary).

Knowledge Check Question

Question: What are other names for acute and chronic AI?

Answer: Acute AI can be called adrenal crisis, or Addisonian crisis. Chronic AI can be called Addison Disease.

How Does Adrenal Insufficiency Present?

Patient symptoms can range from mild, as they are in the chronic form, to life-threatening, as they are in the acute form. The acute form is referred to as adrenal crisis or Addisonian crisis. Symptoms depend on the degree of function loss. Most of the important effects of AI result from a lack of cortisol, but we also see effects from low mineralocorticoid levels and elevated ACTH levels.

Chronic Primary Adrenal Insufficiency

Patients with chronic primary AI (Addison disease) may present very gradually with nonspecific complaints like weakness, fatigue, weight loss, nausea, and muscle aches, reflecting how **low cortisol** impacts general body function.

Depressed mood is common. There may also be orthostatic dizziness as well as mild hyperkalemia and hyponatremia, all caused by the lack of mineralocorticoid hormones that often accompany the low cortisol. Said another way, AI is a cause of hypoaldosteronism, not just hypocortisolism.



Figure 2. Image Credit: ScholarRx, used with permission.

Hyperpigmentation is a distinguishing characteristic of primary AI; it is caused by the high ACTH levels that result from the low adrenal cortisol production.

Hyperpigmentation is a classic sign of Addison disease, and we see it here in a patient with hyperpigmentation of the gums (**Figure 2**).

Why would this occur? Hyperpigmentation occurs in the setting of increased ACTH because the precursor protein to ACTH, proopiomelanocortin (POMC), is cleaved into peptides, which include ACTH and α -melanocyte stimulating hormone (α -MSH). When more ACTH is produced, more α -MSH is produced along with it. Both ACTH and α -MSH can stimulate the production and release of melanin from melanocytes in the skin, leading to hyperpigmentation. Note that this hyperpigmentation does not occur in secondary AI, because ACTH levels (and therefore α -MSH levels) are low.

Knowledge Check Question

Question: Primary adrenal insufficiency leads to Primary Pigmentation of the skin and mucosal surfaces.

Answer: Use the **P** of Primary adrenal insufficiency to remember a key finding.

Secondary Adrenal Insufficiency

In secondary AI there are many of the same symptoms, but hyperpigmentation is absent (because ACTH is low, not high). Also, dizziness, hypotension, hyponatremia, and hyperkalemia are less common because the adrenal gland can still make aldosterone, simulated by the renin-angiotensin system.

Knowledge Check Question

Question: Primary adrenal insufficiency leads to Primary Pigmentation of the skin and mucosal surfaces.



Answer: Use the **P** of **P**imary adrenal insufficiency to remember a key finding.

Adrenal Crisis

In adrenal crisis, many of the previously mentioned symptoms become much more acute and marked. Two important, potentially lethal consequences are hypotension and hypoglycemia; both important to remember with patients in the ICU. Patients in the ICU are at risk for AI for several reasons, including the long-term use of steroids leading to suppression of the adrenals by negative feedback.

Hypotension. Low blood pressure occurs because low cortisol makes the vascular myocytes less sensitive to epinephrine, so they cannot constrict to maintain the blood pressure. The hypotension is typically orthostatic. When primary AI causes adrenal crisis, the low BP is worsened by a concurrent deficiency in the mineralocorticoid aldosterone, which leads to loss of sodium in the kidney that, in turn, reduces extracellular volume.

Hyponatremia. A low serum-sodium concentration is usually caused by an excess of water caused when the posterior pituitary gland secretes too much ADH. Cortisol inhibits ADH secretion, so hypocortisolism leads to increased ADH levels, increased renal water retention, and hyponatremia. A second cause is the renal sodium loss induced by low aldosterone in primary AI; ultimately, low total body sodium leads to hypotension and hyponatremia.

Hypoglycemia. Cortisol is critical in maintaining a normal serum glucose. It does this by increasing gluconeogenesis, increasing peripheral cell insulin resistance, decreasing insulin secretion, and increasing glucagon secretion. It also increases glycogen breakdown into glucose. So when cortisol is low, patients may present with hypoglycemia, especially in children or in adults with concurrent infection. This can cause dizziness, loss of consciousness, or death if severe.

Knowledge Check Question

Question: What are common symptoms resulting from low cortisol?

Answer: Common symptoms include weakness, fatigue, weight loss, hypoglycemia, and orthostatic hypotension.

What Causes Adrenal Insufficiency?

The causes of AI can be divided into primary and secondary. Overall, the most common cause of AI is the abrupt withdrawal of prescribed corticosteroid drugs. (This withdrawal causes secondary AI.) These causes are summarized in **Figure 3**.

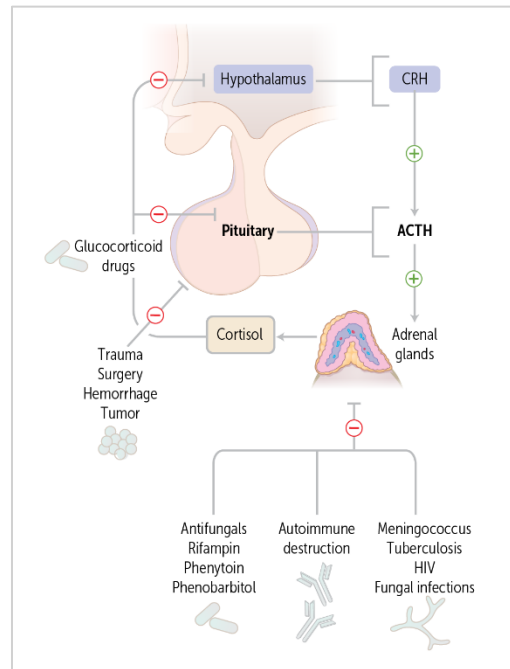


Figure 3. Image Credit: ScholarRx, used with permission.

Primary Adrenal Insufficiency

Primary AI results from the failure or suppression of the adrenal gland; the gland is unable to produce the adrenal hormones cortisol and aldosterone despite normal ACTH stimulation. Primary AI is usually a chronic disorder, and it has several causes.

Autoimmune destruction of the adrenal gland occurs by both cell-mediated immunity and autoantibodies. It occurs mostly in females and may be associated with other autoimmune diseases, like ovarian failure, hypoparathyroidism, type 1 diabetes mellitus, or thyroiditis. The high eosinophil counts (eosinophilia) seen in some patients with primary Addison disease may be part of the autoimmune syndrome that leads to adrenal destruction. In fact, Addison disease is the endocrine abnormality most commonly associated with eosinophilia.

Infections, such as HIV, tuberculosis (TB), and histoplasmosis, can lead to AI. Infections can cause adrenal inflammation (termed adrenalitis), which, in turn, causes AI.



Drugs that inhibit adrenal cortisol synthesis include the antifungals ketoconazole and fluconazole, the anti-TB drug rifampin, the antiparasitic suramin, and the antiseizure drugs phenytoin and phenobarbital.

Clinical Correlation

One example of adrenal crisis is the massive adrenal hemorrhaging that occurs in patients who have sepsis, called Waterhouse-Friderichsen syndrome. The most common cause is a blood stream infection with *Neisseria meningitidis*. We don't completely understand the exact cause of the hemorrhage. It may be that bacterial seeding within the adrenal gland leads to dysfunction of the blood clotting mechanisms and consequent hemorrhage.

Knowledge Check Question

Question: What is the pathology in primary AI?

Answer: The adrenal glands do not produce enough endogenous steroids to meet the basic metabolic demands of the body.

Secondary Adrenal Insufficiency

Secondary AI results from decreased anterior pituitary ACTH release. It does not cause hyperpigmentation because ACTH is low. It also does not usually cause hyperkalemia or metabolic acidosis because the adrenal ability to make aldosterone is preserved. This is because the primary regulator of aldosterone is the renin-angiotensin-aldosterone system, which is still intact. The most common cause of secondary AI is the abrupt withdrawal of glucocorticoids, although there are other causes as well.

Withdrawal of Glucocorticoids. The abrupt withdrawal from prolonged use of glucocorticoid drugs like prednisone is the most common cause of **AI overall**. The problem is not taking the drug, but what happens when the patient stops it abruptly (rather than slowly tapering it off).

Glucocorticoid drugs act like endogenous cortisol to suppress the HPA axis (lower CRH and ACTH secretion). Over time, the low ACTH levels lead to adrenal atrophy and insufficiency (secondary AI), so the gland cannot make cortisol. As long as patients take the glucocorticoid drug, they will appear normal or have signs of hypercortisolism (Cushing syndrome) like moon facies and mood swings. But, if the drug is abruptly stopped, the suppressed ACTH secretion and insufficient adrenal gland cannot produce



enough cortisol to maintain body functions, and signs of hypocortisolism can occur, including nausea, vomiting, hypotension, and hypoglycemia.

Clinical Correlation

Patients who have used glucocorticoid drugs for a few weeks should always be instructed to gradually taper off the drugs to avoid adrenal crisis.

Pituitary Insufficiency. This is a much less common cause than glucocorticoid withdrawal. Pituitary disease results in the deficiency of one or more pituitary hormones, which, in turn, can cause low ACTH release. This disease is most commonly caused by head trauma, pituitary tumors, brain surgery, and Sheehan syndrome, a condition characterized by pituitary necrosis that can follow a postpartum hemorrhage. This necrosis may be accompanied by deficiencies in other pituitary hormones like growth hormone, thyroid-stimulating hormone, or prolactin.

How Do We Diagnose Adrenal Insufficiency?

Diagnosis of AI occurs in two steps. First, we document low cortisol secretion. Then, we determine the source of the problem.

Document Low Cortisol Secretion

Diagnosis of AI starts with measurement of morning serum cortisol concentration. Normal cortisol levels range from roughly 5 to 26 mcg/dL. Usually, serum cortisol is highest in the morning. So a low concentration of serum cortisol early in the morning strongly suggests AI.

An **ACTH stimulation test** can be done if the morning cortisol result is low or inconclusive. Synthetic ACTH is given and the cortisol level checked before and after the ACTH test. A twofold increase in cortisol is normal. A lesser response to ACTH (0%-25%) most often suggests primary AI.

Clinical Correlation

This stimulation test is often done in the ICU to test for AI in patients with hypotension.

Why? Because primary AI is so common in patients in the ICU. Because this is an emergency, patients with suspected adrenal crisis are given cortisol immediately, then can be tested later.



Determine the Cause

Both the ACTH stimulation test and the plasma ACTH concentration are useful here.

Primary Adrenal Insufficiency. The basal ACTH level is high and the ACTH stimulation test shows a <25% increase in cortisol levels. Patients should then receive a CT scan of the adrenals to look for signs of atrophy, hemorrhage, or infection.

Secondary Adrenal Insufficiency. The basal ACTH level is low, and the ACTH stimulation test shows a variable response. If there has been long-term low ACTH secretion, the adrenal gland may be atrophied and not respond to ACTH, so the response can look like primary hypocortisolism (0%-25% response). On the other hand, if the ACTH secretion has dropped acutely, the adrenal gland will respond briskly to ACTH stimulation, with a 10- to 15-fold increase in cortisol, more than the normal twofold increase.

This table summarizes these lab findings.

	Morning Serum Cortisol	Plasma ACTH Concentration	ACTH Stimulation Test
Primary AI	Low	High	Low cortisol response
Secondary AI	Low	Low	Varies: low to high

Knowledge Check Question

Question: How is the ACTH level used in diagnosing AI?

Answer: The ACTH level is high in primary insufficiency, low in secondary.

Additional Tests

Patients with primary insufficiency should be tested for HIV, because this is a common and treatable cause. Testing for TB or fungal infections may be appropriate depending on the presentation.

In secondary insufficiency, patients should be tested for other hormone deficiencies like prolactin, insulin-like growth factor-1 (used for growth hormone assessment), and thyroid-stimulating hormone (TSH).

Differential Diagnosis

Many of the symptoms of AI are vague. In fact, some of them, such as fatigue and hypoglycemia, may overlap with symptoms of studying for too long. That said, it is important to keep your differential diagnosis broad until you have reasons to rule in or out a diagnosis. A few important considerations for your differential are listed here.

Fatigue has a wide differential diagnosis, including depression, thyroid disease, and many systemic illnesses and infections.

Hypotension can be caused by sepsis, dehydration, or autonomic nerve dysfunction.

Hypoglycemia can be caused by drugs (insulin, oral antihyperglycemic agents), infection, or metabolic disorders (e.g., reactive hypoglycemia).

How Do We Manage and Prevent Adrenal Insufficiency?

Treatment is designed to help patients manage symptoms and involves hormone replacement therapy.

Glucocorticoid Hormones

AI is treated with glucocorticoid hormones. These are available in a variety of IV and oral forms (e.g., cortisone, dexamethasone, prednisone).

The dose should be increased during stressful events (e.g., trauma, surgery, or severe illness), as stress can trigger an Addisonian crisis in patients with Addison disease. Any signs of infection (e.g., fever) should prompt a call to the clinician, because such stresses may cause adrenal crisis, and their cortisol dose may need to be increased. Also, these patients may require higher drug doses during other acute illnesses like infections because the stress of these infections creates higher cortisol demands. Additionally, the glucocorticoids may cause immunosuppression and lead to infections.

In severe cases of acute AI, the glucocorticoids are given through an IV line. If needed, IV saline is added to raise blood pressure, and IV dextrose is added to correct hypoglycemia.

If the patient also has hyperkalemia and hyponatremia, a combination that suggests mineralocorticoid deficiency, they can be given fludrocortisone, a potent synthetic mineralocorticoid that acts like aldosterone.

Prevention

To prevent AI resulting from glucocorticoid withdrawal, patients using high-dose glucocorticoid drugs for a prolonged period of time should undergo a drug taper at the end of their treatment. The schedule should be carefully written out, and the patient should be coached on how to follow it precisely.

Knowledge Check Question

Question: Why is fever of concern in patients taking glucocorticoid drugs?

Answer: Immunosuppression caused by the drugs may lead to increased infections, and infection may precipitate adrenal crisis.

Case Connection

Thinking back to HK, what tests are most appropriate?

After seeing her clinician, HK calls and tells you that her first tests showed low sodium and blood sugar levels and high potassium levels. Her clinician just called and said that her cortisol level was low, too. She thinks HK has Addison disease, and she has an ACTH test scheduled. HK is worried about taking corticosteroid drugs (she has heard that they cause weight gain), but she is glad to have an explanation for her symptoms.

Summary

- Adrenal insufficiency (AI) is an inability of the adrenal glands to produce enough glucocorticoids or mineralocorticoids (or both) to meet the metabolic demands of the body.
- AI can be classified as primary, resulting from adrenal disease; or secondary, because of a lack of ACTH secretion by the pituitary gland.
- The most common type of AI is secondary AI because of glucocorticoid drug withdrawal.
- Adrenal crisis occurs when body stressors cause the need for cortisol to exceed the supply, leading to severe illness.
- Common symptoms of AI include weakness, fatigue, weight loss, muscle aches, orthostatic hypotension, and hypoglycemia.
- Primary AI, but not secondary AI, can cause hyperpigmentation of the skin and hyperkalemia.
- The most common cause of secondary AI is the abrupt withdrawal of glucocorticoid drugs; the abrupt withdrawal ultimately causes low cortisol levels.
- Causes of primary AI are infection, drugs, autoimmune adrenal destruction, and adrenal hemorrhage.
- Besides exogenous corticosteroids, common causes of secondary AI are pituitary tumors, brain surgery, and head trauma.
- AI should be evaluated first by measuring the concentration of early morning serum cortisol.



- If the morning serum cortisol concentration is low, it should be followed by measurement of plasma ACTH concentration and ACTH stimulation testing to distinguish between primary and secondary AI.
- AI is treated with glucocorticoid drugs and, if needed, saline, glucose, and fludrocortisone.
- To prevent AI resulting from glucocorticoid withdrawal, patients using high-dose glucocorticoid drugs for a prolonged period of time should undergo a drug taper at the end of their treatment.
- To prevent acute adrenal crisis in patients with Addison disease, glucocorticoid dosing is increased for stressful events.

Check Your Knowledge

Knowledge Check Question

*Question: An otherwise-healthy 19-year-old male presents to the emergency department for worsening headache, a stiff neck, and a high fever. Cerebrospinal fluid cultures show *Neisseria meningitidis*. On examination, he is found to be hypotensive, and laboratory results show hyperkalemia and metabolic acidosis. Which of the following endocrine disorders is most likely?*

Answer:

- **Cushing syndrome. Incorrect.** This is caused by increased production of cortisol. Symptoms include hypertension, weight gain, moon facies, abdominal striae, and truncal obesity.
- **Exogenous steroid withdrawal. Incorrect.** Exogenous steroid withdrawal would produce similar symptoms; however, this patient has no medical history of, and no risk factors for, exogenous steroid use.
- **Hypothyroidism. Incorrect.** Hypothyroidism presents with symptoms of cold intolerance, weight gain, decreased appetite, hypoactivity, constipation, and decreased reflexes, but it is typically more insidious in nature and is not associated with meningitis infections.
- **Pheochromocytoma. Incorrect.** This is a rare tumor that secretes catecholamines; it presents with recurrent episodes of increased blood pressure, headache pains, increased perspiration, palpitations, and pallor.
- **Waterhouse-Friderichsen syndrome. Correct.** The patient is presenting with signs and symptoms of primary acute adrenal insufficiency (AI) secondary to an N meningitidis infection. Waterhouse-Friderichsen syndrome is an acute primary AI caused by massive hemorrhage in the adrenal glands.

Part 2 of 6: Hyperaldosteronism

Overview

Learning Objectives

This page covers the following learning objectives:

4. Understand the pathogenesis and clinical features of Conn's Syndrome.
6. Define Conn's syndrome (primary hyperaldosteronism) and describe the underlying pathophysiology, including aldosterone-producing adrenal adenomas and bilateral adrenal hyperplasia.

What Is Hyperaldosteronism?

Case Connection

BY is a 44-year-old construction worker with hypertension, for which he takes chlorthalidone, potassium, and verapamil. Despite BY's adherence to his medication regimen, his blood pressure (BP) was high at the pharmacy kiosk twice last week, and he reports intermittent headaches. His BP today is 159/92 mm Hg, and his serum potassium level is low. You are concerned about an underlying problem: the BP medications should be working, and the potassium supplement should be keeping his potassium levels in check.

What will you recommend to BY as the next steps in his treatment? Consider your answer as you read, and we'll revisit BY at the end of the brick.

Hyperaldosteronism is a condition in which the adrenal cortex produces excessive aldosterone. It's an important cause of secondary hypertension—high blood pressure (BP) due to an underlying cause. That's important to diagnose—correcting the hyperaldosteronism will bring the BP back to normal, and the patient can avoid a lifetime of BP medications.

Aldosterone, which is a mineralocorticoid hormone made in the adrenal glands, plays a central role in the regulation of BP and blood volume. Aldosterone stimulates reabsorption of renal sodium and water to help maintain fluid balance. Its other main function is to promote excretion of renal potassium and hydrogen ions.

Hyperaldosteronism is classified as either primary or secondary:



- **Primary** hyperaldosteronism, or Conn syndrome, occurs when the adrenal glands act on their own to produce too much aldosterone. It is rare.
- **Secondary** hyperaldosteronism occurs when a source outside the adrenal glands causes the adrenal cortex to release excess aldosterone. This form is more common than primary. Common causes include ascites, left-sided heart failure, and pulmonary hypertension, all of which decrease cardiac output; the decrease in blood volume results in an increase in renin, leading to a secondary increase in aldosterone.

Knowledge Check Question

Question: Is primary or secondary hyperaldosteronism more common?

Answer: Secondary hyperaldosteronism is more common.

How Do Patients With Hyperaldosteronism Present?

Patients are usually asymptomatic when they have hyperaldosteronism. Asymptomatic findings likely to show up on laboratory exam include hypertension, hypokalemia, and metabolic alkalosis (discussed below).

That said, there is also a wide range of symptoms for hyperaldosteronism. These could include muscle spasm, muscle weakness, fatigue from hypokalemia, or headache or chest pain if there is very severe (emergent) hypertension.

Knowledge Check Question

Question: What are some common symptoms of hyperaldosteronism?

Answer: When present, symptoms of hyperaldosteronism can include muscle cramps, weakness, fatigue, headache, or chest pain.

What Is the Pathophysiology of Hyperaldosteronism?

Primary and secondary hyperaldosteronism have different causes.

Primary Hyperaldosteronism

There are three main causes of primary hyperaldosteronism. The two most common causes are adrenal adenoma and bilateral adrenal hyperplasia. Unilateral adrenal hyperplasia is rare. Let's look a little closer at each cause.

Adrenal Adenoma. This is a benign neoplasm of the zona glomerulosa, the outermost layer of the adrenal cortex (**Figure 1**).

It may secrete one or more adrenal cortical hormones, including aldosterone, cortisol (causing Cushing syndrome), androgens (causing virilizing syndromes such as facial hair in females), or a mix of these.



Figure 1. Image Credit: ScholarRx, used with permission.

Bilateral Adrenal Hyperplasia. This condition is a benign enlargement of both adrenal glands, most often a diffuse increase in the number of adrenocortical cells. But it can sometimes be micronodular instead. Note that this is not the same thing as **congenital adrenal hyperplasia**, which is due to enzyme deficiencies and presents in infancy or childhood.

Rare Genetic Abnormalities. Included in this third group of causes are two types of **congenital adrenal hyperplasia**, 11 β -hydroxylase deficiency and 17 α -hydroxylase deficiency, which cause increased mineralocorticoids and hypertension. In these disorders, cortisol synthesis is blocked, and the adrenal synthetic pathway can increase synthesis of mineralocorticoids, leading to hypertension and hypokalemia in children.

Another rare genetic abnormality is **aldosterone synthase hyperactivity**, in which a mutation results in an aldosterone synthase enzyme that is sensitive to ACTH, resulting in hyperactive aldosterone synthase and high levels of aldosterone. Giving cortisol to this last group suppresses the ACTH and relieves the problem. This is why it is sometimes called glucocorticoid-remediable hyperaldosteronism.

Knowledge Check Question

Question: What are the most common causes of primary hyperaldosteronism?

Answer: Adrenal adenoma and bilateral adrenal hyperplasia are the most common causes of primary hyperaldosteronism.

Secondary Hyperaldosteronism

Secondary hyperaldosteronism results from overactivation of the **renin-angiotensin-aldosterone system (RAAS)**, which is the main regulator of aldosterone. Specifically, anything that increases renin levels will then increase angiotensin II and then aldosterone. The following conditions will increase renin levels:

Renal artery stenosis: This condition impairs blood flow through the renal arteries to the renal juxtaglomerular cells adjacent to the afferent arteriole, stimulating their release of renin. It's the most common cause of secondary hyperaldosteronism and is a common cause of secondary hypertension.

Edematous disorders: Conditions like congestive heart failure, cirrhosis, chronic kidney disease, and nephrotic syndrome cause low renal blood flow and sometimes increase renin and aldosterone levels. They must be ruled out before launching off into a detailed hormonal workup of hyperaldosteronism. Note that high aldosterone does not cause edema (see below) but that these low blood flow disorders associated with edema cause high aldosterone levels.

Juxtaglomerular tumors: They produce renin and are a rare cause of high aldosterone levels.

Complications

Patients with hyperaldosteronism are usually asymptomatic; usually, the main consequences of hyperaldosteronism first turn up on the physical or lab exams. These effects include hypertension and the serum electrolyte abnormalities such as hypokalemia and metabolic alkalosis.

Hypertension. Hyperaldosteronism is a common cause of secondary hypertension (i.e., hypertension that has a clear, identifiable cause). Patients often present with new onset high BP, very high BP (including emergent hypertension of >180/120 mm Hg with complications such as chest pain or stroke), or BP that is difficult to control with routine antihypertensive drugs. Because aldosterone increases renal sodium and water reabsorption, volume expansion accounts for at least part of the hypertension seen in

patients with hyperaldosteronism. Interestingly, the patients generally do not appear volume expanded or edematous.

Why not? The answer is the aldosterone escape mechanism. In this mechanism, the initial volume expansion triggers the secretion of atrial natriuretic peptide (ANP). ANP is a peptide hormone that is synthesized and secreted by atrial cardiac myocytes in response to volume expansion, causing stretching of the atrial wall. ANP promotes renal sodium (and water) loss and counterbalances the effect of aldosterone to increase blood volume. So, while there may still be hypertension, the patient usually does not appear edematous.

Electrolyte Abnormalities. Usually, high BP is the first sign of hyperaldosteronism, and the second is concurrent **hypokalemia** and **metabolic alkalosis** on lab testing results. This combination of lab findings is not typical of patients with hypertension unless they are already taking a diuretic drug. Hypokalemia results from the urine potassium losses directly caused by aldosterone.

The hypokalemia can occasionally be severe enough to cause muscle cramping or weakness. ECG changes like **T-wave inversion** and **prominent U waves** may sometimes be seen (**Figure 2**).

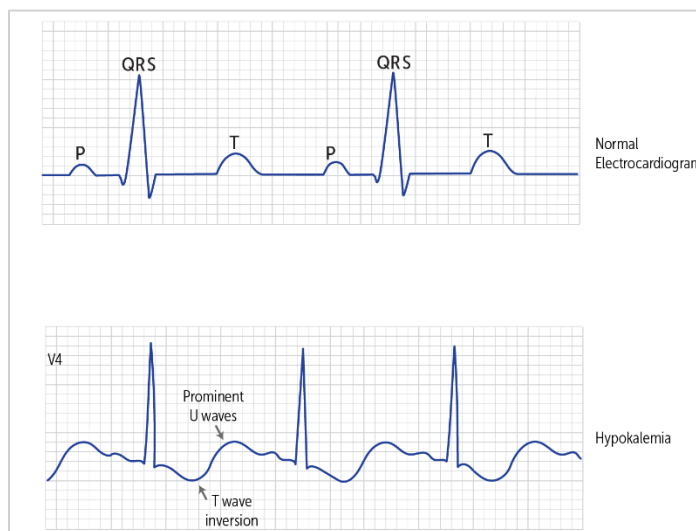


Figure 2. Image Credit: ScholarRx, used with permission.

Metabolic alkalosis (increase in blood pH) occurs because of hydrogen ion loss in the urine. Unlike the most common causes of metabolic alkalosis like vomiting or diuretics, this type cannot be corrected with volume expansion because the problem is due to excess aldosterone that causes continuing hydrogen ion losses in the urine. The hypokalemia also contributes to the metabolic alkalosis because of increased H^+-K^+ transcellular exchange. (K^+ is reclaimed at the expense of H^+ excretion.)

Knowledge Check Question

Question: How does hyperaldosteronism lead to hypertension?

Answer: Excessive aldosterone levels increase renal reabsorption of sodium and water.

How Do We Diagnose Hyperaldosteronism?

The diagnostic workup usually happens when a patient has new or worsening hypertension and also has hypokalemia and/or metabolic alkalosis. If this is the case, the next thing to do is document that aldosterone levels are high and then figure out why.

Laboratory Studies

First comes the concurrent measurement of **plasma aldosterone** and **plasma renin activity (PRA)**, often expressed as a ratio. The normal value of this ratio is about 10 (when aldosterone concentrations are expressed as ng/dL). Primary hyperaldosteronism shows increased levels of aldosterone with decreased levels of PRA. This is because the high aldosterone levels produced by the adrenal gland suppress renin activity.

- Primary hyperaldosteronism usually shows aldosterone-to-renin ratios of >30:1.
- Secondary hyperaldosteronism caused by high renin levels (eg, renal artery stenosis) shows both high PRA and high aldosterone. The aldosterone-to-renin ratio should then be near normal (10:1).

When undergoing these tests, the patient should not be volume depleted, which would itself increase the renin and aldosterone levels and confuse the diagnosis. So, commonly, the patient is encouraged to eat salty snacks and drink fluids before the test.

If there is doubt, the **aldosterone suppression test** is used to confirm the diagnosis. It's done by giving the patient oral or intravenous sodium chloride followed by renin and aldosterone measurement. Normal patients will lower their renin and aldosterone levels, but patients with hyperaldosteronism will not.

A comparison of lab results for primary and secondary hyperaldosteronism shows there is only one difference between the two—the renin activity.



Finding	Primary Hyperaldosteronism	Secondary Hyperaldosteronism
Serum aldosterone level	High	High
Plasma renin activity	Low	High
Serum potassium level	Low	Low
Serum sodium level	Increased (mildly)	Increased (mildly)
Metabolic alkalosis	Yes	Yes

Radiology Studies

If the patient's lab results indicate primary hyperaldosteronism, a CT scan of the adrenal glands is done to look for tumors or bilateral adrenal hyperplasia. If there is uncertainty, sampling the renal vein for aldosterone can be done to see if the aldosterone is elevated on just one side (adenoma) or on both sides (bilateral adrenal hyperplasia).

If secondary hyperaldosteronism is detected, renal artery stenosis is evaluated by CT angiography or Doppler sonography. Figure 3 shows bilateral renal artery stenosis (arrows).



Figure 3. Image Credit: ScholarRx, used with permission.

Differential Diagnosis

Hyperaldosteronism is a type of secondary hypertension. Recall that secondary hypertension accounts for only 5% of patients with hypertension—the rest have essential (primary) hypertension. Below lists other causes of secondary hypertension, from most to least common.



Cause	Findings	Etiology
Chronic kidney disease	Edema, elevated serum creatinine	Glomerular or interstitial disease
Renal artery stenosis	Abdominal or flank bruit, hypokalemia	Atherosclerosis, fibromuscular dysplasia
Hyperaldosteronism	Hypokalemia	Adrenal adenoma, adrenal hyperplasia
Cushing syndrome	Buffalo hump, moon facies, striae	Prescribed corticosteroid drugs, adrenal or pituitary tumor
Pheochromocytoma	Tachycardia, sweating, palpitations, headache	Chromaffin cell neoplasia

Knowledge Check Question

Question: Is low renin activity with high aldosterone characteristic of primary or secondary hyperaldosteronism?

Answer: Low renin activity with high aldosterone is characteristic of primary hyperaldosteronism.

How Do We Manage Hyperaldosteronism?

There are different treatment options, depending on what is causing the excess aldosterone.

Surgery

For patients with primary hyperaldosteronism due to adrenal adenoma, the best treatment is laparoscopic removal of the affected adrenal gland (**laparoscopic adrenalectomy**). About half of patients will have their hypertension cured by this procedure.

Secondary hyperaldosteronism caused by renal artery stenosis is treated by either angiotensin-converting enzyme (ACE) inhibitor therapy or angioplasty of the renal artery.

Medication

For primary hyperaldosteronism due to bilateral adrenal hyperplasia, the best treatment is medication. The medications of choice are the K⁺-sparing diuretics **spironolactone** and **eplerenone**, both of which are competitive aldosterone receptor antagonists. These

drugs bind to aldosterone receptors in the adrenal cortex, decreasing the amount of aldosterone that can bind to the receptors and hence reducing the hormone's effects. Medications are preferred for this condition, since removing both adrenal glands is an extreme therapy that would then require the patient to take pharmacologic glucocorticoids and synthetic mineralocorticoids for life.

Because spironolactone is also an antagonist of the androgen receptor and an agonist of the progesterone receptor, some patients (both male and female) may get painful breast swelling (gynecomastia). This can be reduced by using the alternative agent eplerenone.

Knowledge Check Question

Question: What potassium-sparing diuretics are used for hyperaldosteronism?

Answer: ironolactone and eplerenone are used for hyperaldosteronism.

Case Connection

Thinking back to BY, what do you recommend as the next steps in his treatment? In your workup for secondary causes of hypertension, BY's aldosterone level is high, but his aldosterone to renin ratio is normal. You explain that something might be wrong with the blood flow to his kidneys, and you order Doppler ultrasonography to evaluate for renal artery stenosis. The sonogram shows a stenosis in the right renal artery. You reassure BY that this condition can likely be corrected by angioplasty of that artery and refer him to an interventional radiologist to correct the problem.

Summary

- Hyperaldosteronism is a condition in which the adrenal glands produce excessive aldosterone levels; it can be primary or secondary.
- Patients are often asymptomatic; when patients are symptomatic, they may present with muscle cramps, weakness, fatigue, and headaches.
- Hyperaldosteronism causes secondary hypertension, hypokalemia, and metabolic alkalosis.
- Primary hyperaldosteronism is caused most often by adrenal hyperplasia or adrenal adenoma.
- Secondary hyperaldosteronism is most often caused by renal artery stenosis with high renin levels.



- Hyperaldosteronism is best diagnosed by measuring serum aldosterone and renin levels.
- The renin level will distinguish between primary (low renin) and secondary (high renin) types.
- Adenomas can be resected endoscopically.
- Renal artery stenosis is treated with angiotensin-converting enzyme inhibitors or renal artery angioplasty.
- Bilateral adrenal hyperplasia is treated with competitive aldosterone receptor antagonists, such as spironolactone and eplerenone.

Check Your Knowledge

Knowledge Check Question

Question: Which of the following is most often observed in patients with hyperaldosteronism?

Answer:

- **Edema. Incorrect.** This is not typical, even though sodium reabsorption is increased in the kidney; the atrial natriuretic peptides counteract aldosterone effects by promoting renal sodium loss.
- **Hypokalemia. Correct.** This results from the action of aldosterone on the collecting duct, causing potassium loss in the urine. Other effects are hypertension and metabolic alkalosis.
- **Hypotension. Incorrect.** This is not seen; hypertension is common instead, partly because of sodium and water reabsorption.
- **Hypermagnesemia. Incorrect.** Hypomagnesemia, not hypermagnesemia, is sometimes seen.
- **Metabolic acidosis. Incorrect.** This is not seen in patients with hyperaldosteronism. Instead, metabolic alkalosis occurs from increased urinary loss of H^+ and K^+ resulting from excessive levels of aldosterone.

Part 3 of 6: Hypoaldosteronism

Overview

Learning Objectives

This page covers the following learning objectives:

6. Define Conn's syndrome (primary hyperaldosteronism) and describe the underlying pathophysiology, including aldosterone-producing adrenal adenomas and bilateral adrenal hyperplasia.

What Is Hypoaldosteronism?

Case Connection

TR is a 56-year-old male who established care with you 1 month ago, 2 weeks after his prior clinician retired. On reviewing TR's chart, you noted that his electrolytes and renal function were normal but that his blood pressure had not been well controlled on a low-dose thiazide diuretic. On his first visit with you, you added lisinopril, an angiotensin-converting enzyme (ACE) inhibitor. During today's follow-up, TR's blood pressure has improved, but he complains of weakness and palpitations. You check a routine metabolic panel. TR's potassium is 5.8 mEq/dL (normal, 3.5-5.2 mEq/dL.)

What could be causing TR's elevated potassium? What is the etiology of this clinical condition? What other diagnostic testing would you order? Consider your answers as you read, and we'll revisit TR at the end of the brick.

Hypoaldosteronism is a condition in which the adrenal cortex produces too little aldosterone. It often accompanies cortisol deficiency in Addison disease, but can occur on its own. It also can be a complication of diabetes mellitus caused by low renin secretion. In any case, the consequences can be low blood pressure and hyperkalemia.

Aldosterone, which is a mineralocorticoid hormone made in the adrenal glands, plays a central role in the regulation of blood pressure and blood volume. Aldosterone stimulates sodium and water reabsorption to help maintain fluid balance. Its other main function is to promote renal potassium and hydrogen ion secretion. The consequences of hypoaldosteronism are easy to predict if you know these normal actions. There are two main types to consider:



- **Primary hypoaldosteronism** occurs when the adrenal cortex itself is unable to produce aldosterone, even in the setting of elevated renin stimulation. Primary aldosterone deficiency is usually caused by adrenal insufficiency (Addison disease) or rare deficiencies of enzymes such as 21-hydroxylase or aldosterone synthase.
- **Secondary hypoaldosteronism** is more common than primary. It occurs when the adrenal cortex is functioning normally, but there is lack of stimulation by renin, usually caused by an impairment of the renin-angiotensin-aldosterone system (RAAS). Diabetes mellitus is a common cause.

Knowledge Check Question

Question: Aldosterone plays a key role in regulating which body functions?

Answer: Aldosterone helps regulate blood pressure, blood volume, potassium excretion, and hydrogen ion excretion.

How Do Patients With Hypoaldosteronism Present?

Regardless of whether the cause is primary or secondary, patients with low aldosterone can experience **dizziness and orthostatic hypotension**, because they have low renal sodium reabsorption and low extracellular volume. Their potassium and hydrogen ion excess may lead to **hyperkalemia** and **metabolic acidosis**.

These serum abnormalities are usually asymptomatic. However, if the potassium excess becomes severe enough, patients may have muscle weakness, palpitations, or even cardiac arrest—hyperkalemia directly affects the heart.

Knowledge Check Question

Question: What does hypoaldosteronism do to serum potassium and urine sodium?

Answer: Hypoaldosteronism increases serum potassium and urinary sodium.

What Is the Pathophysiology of Hypoaldosteronism?

As stated above, primary hypoaldosteronism is caused by adrenal gland abnormalities, whereas secondary hypoaldosteronism is caused by low RAAS activity or certain medications. The secondary form is much more common.

Primary Hypoaldosteronism

Primary hypoaldosteronism arises in the adrenal gland itself. This can lead to combined deficiencies of adrenal hormones (eg, cortisol and aldosterone). The following conditions are the main causes of primary hypoaldosteronism:

- **Primary adrenal insufficiency** (Addison disease): This condition will usually reduce both cortisol and aldosterone synthesis, leading to the findings of hypoaldosteronism. Because both cortisol and aldosterone support the blood pressure, these patients can be very hypotensive. Primary adrenal insufficiency can be caused by autoimmune destruction, infection, or drugs.
- **Severe illness:** Severe illness can also reduce aldosterone release; this situation is seen a lot in the intensive care unit.
- **21 α -Hydroxylase deficiency:** This form of congenital adrenal hyperplasia will reduce aldosterone as well as cortisol. It presents in children.
- **Genetic causes:** Rare inherited disorders of aldosterone synthesis exist, but we will not detail those here.

Secondary Hypoaldosteronism

Secondary hypoaldosteronism is generally caused by abnormally low activity of the renin-angiotensin-aldosterone system (RAAS), which is the main stimulus of aldosterone secretion. Low RAAS activity can come from low levels of renin or angiotensin II.

Renin is an enzyme released by the renal juxtaglomerular (JG) cells. It converts the proenzyme angiotensinogen to angiotensin I, which is in turn converted to angiotensin II by ACE (**Figure 1**).

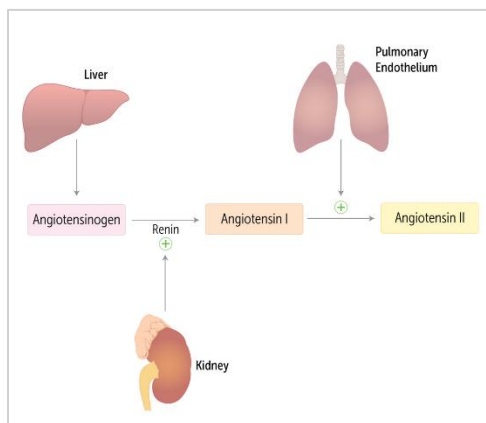


Figure 1. Image Credit: ScholarRx, used with permission.



Angiotensin II has direct effects to constrict blood vessels, but also stimulates aldosterone secretion by the adrenal cortex. The general function of the system is to maintain blood pressure along with extracellular volume and blood volume. Therefore, anything that decreases renin secretion or decreases angiotensin II will secondarily decrease aldosterone secretion. Let's look at these causes one at a time.

Low Renin. So what causes persistent **low renin excretion**, which we call **hyporeninemic hypoaldosteronism**?

- **Chronic kidney disease (CKD):** When the renin-secreting JG cells become damaged, secondary hypoaldosteronism results. CKD is most commonly caused by diabetes and hypertension. Other causes are chronic urinary tract obstruction and interstitial renal disease, which are often a result of chronic nonsteroidal anti-inflammatory drug (NSAID) use.
- **Diabetes mellitus:** This is another common cause of secondary hypoaldosteronism. This may be in part because of **diabetic autonomic neuropathy**. The sympathetic nerves normally stimulate the JG cells to make renin in response to low blood pressure or blood volume, and if the nerves are damaged by diabetes, renin production drops. This complication of diabetes is seen most often in patients with diabetic nephropathy, the glomerular disease associated with longstanding diabetes.
- **Drugs:** NSAIDs as well as the antirejection drugs cyclosporine and tacrolimus may reduce renin secretion.

Low Angiotensin II. Low angiotensin II (ATII) levels (with normal or high renin) are most commonly caused by **ACE inhibitors**. By inhibiting angiotensin-converting enzyme (ACE), these drugs reduce conversion of ATI to ATII. The ability to lower ATII makes the drugs good for blood pressure control and for kidney protection, as lower ATII levels reduce both arteriolar constriction and intrarenal capillary pressure. But the low ATII levels will also reduce aldosterone levels, causing hyperkalemia in some cases. **Hyperkalemia** is a common side effect of these drugs.

Low Adrenal Aldosterone Secretion or Effect. Low adrenal aldosterone secretion or effect (with normal renin and ATII and a normal adrenal gland) is largely caused by drugs:

- Most common are drugs that reduce aldosterone release. These include heparin, NSAIDs, and angiotensin-receptor blockers.
- Aldosterone receptor antagonists such as spironolactone don't reduce aldosterone levels but do block its effect.

Knowledge Check Question

Question: ACE inhibitors block the formation of what peptide?

Answer: ACE inhibitors block the formation of angiotensin II.

Complications

Regardless of cause, patients with hypoaldosteronism are at risk for **hyperkalemia** and **volume depletion** resulting from the limited action of aldosterone. The hyperkalemia can be severe and can cause cardiac abnormalities if potassium levels greater than 5.5 to 6.0 mEq/L.

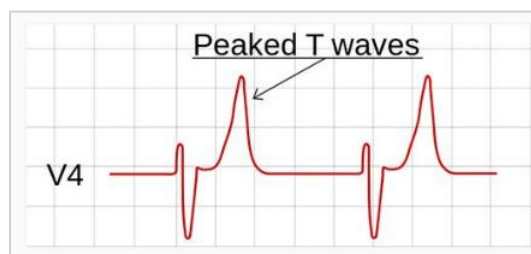


Figure 2. Image Credit: ScholarRx, used with permission.

The best warning sign for this are peaked T waves on ECG (**Figure 2**). It's important to intervene immediately if you see this. If potassium levels exceed 8 mEq/L, lethal arrhythmias can occur.

Another complication of hypoaldosteronism is **type IV (hyperkalemic) renal tubular acidosis**, a type of metabolic acidosis with a normal anion gap. Here, the low aldosterone level causes chronic hyperkalemia, which, in turn, inhibits the production of ammonium by the renal proximal tubule. Ammonium acts as an important buffer for renal acid excretion, so a low level of renal ammonia production means a low rate of acid excretion and the development of a chronic metabolic acidosis.

Knowledge Check Question

Question: How does hyperkalemia affect ECG?

Answer: Hyperkalemia makes peaked T waves.



How Do We Diagnose Hypoaldosteronism?

The diagnosis is most often suspected when patients have hyperkalemia and a normal-anion gap metabolic acidosis. Usually the history, physical exam, and routine labs (eg, serum potassium, bicarbonate, and creatinine) will identify one of the causes listed above, because drugs and kidney disease are the two main causes.

Measuring serum aldosterone and renin levels usually is not warranted. Primary adrenal insufficiency is ruled out by checking a random serum cortisol level and, if needed, with adrenocorticotrophic hormone stimulation testing.

Differential Diagnosis

Most patients who present with dizziness and orthostatic hypotension have simple volume depletion caused by vomiting, lack of fluid intake, diarrhea, or heat exposure. Older adult patients and patients with diabetes mellitus may have dysregulated sympathetic nervous system function, causing orthostatic hypotension. Note that among these patients, only those with diabetes would also be likely to have hyperkalemia, as explained above. In addition, patients with Addison disease, or adrenal insufficiency, may have hypoaldosteronism along with their low cortisol.

Knowledge Check Question

Question: A deficiency of which other hormone often accompanies low aldosterone?

Answer: Cortisol deficiency often accompanies low aldosterone.

How Do We Manage Hypoaldosteronism?

For hyperkalemia, treatment consists of removing offending drugs if hyperkalemia is severe—serum potassium greater than 5.5 mg/dL or ECG changes such as peaked T waves.

Regarding volume depletion, if the patient has hypotension or dizziness, the problem may be corrected by increasing sodium in the patient's diet or, in hospital patients, administering normal saline. Chronically, administering the synthetic mineralocorticoid fludrocortisone may be helpful, but it has the potential adverse effect of excess sodium retention, causing edema.



Knowledge Check Question

Question: What synthetic mineralocorticoid may be administered to treat hypoaldosteronism?

Answer: Fludrocortisone may be administered to treat hypoaldosteronism.

Case Connection

Thinking back to TR, what could be causing his elevated potassium? What is the etiology of this condition? What other diagnostic testing would you order? You suspect ACE inhibitor-induced hypoaldosteronism as the cause of TR's elevated potassium, and you begin your clinical reasoning by first reviewing in your head normal renal physiology: ACE converts angiotensin I to angiotensin II, and angiotensin II stimulates release of aldosterone. Because lisinopril is an ACE inhibitor, angiotensin II and aldosterone levels fall. The resultant hypoaldosteronism leads to reduced renal potassium excretion and an elevated serum potassium. Concerned about hyperkalemia's effects on the heart, you check an ECG, which shows peaked T waves, consistent with severe hyperkalemia.

After appropriate acute treatment for TR's hyperkalemia, his potassium level normalizes. You stop the ACE inhibitor and substitute another antihypertensive medication, and his blood pressure is well controlled at his next visit.

Summary

- Hypoaldosteronism is a condition in which aldosterone levels are abnormally low; it can be primary (caused by adrenal disease) or secondary (caused by abnormalities outside the adrenal glands).
- Hypoaldosteronism often presents with dizziness and orthostasis (postural hypotension).
- Primary hypoaldosteronism is most often caused by primary adrenal insufficiency (Addison disease).
- Secondary hypoaldosteronism is most often caused by diabetes mellitus, chronic kidney disease (low renin), or angiotensin-converting enzyme inhibitors (low angiotensin II).
- Hypoaldosteronism may cause volume depletion, hyperkalemia, and type IV (hyperkalemic) renal tubular acidosis.



- Hypoaldosteronism can be diagnosed by history and a physical along with routine labs showing hyperkalemia and normal anion gap metabolic acidosis.
- Treatment of hypoaldosteronism includes removal of the offending agent or treatment of the underlying medical condition.
- The synthetic mineralocorticoid fludrocortisone may also be used to treat hypoaldosteronism.

Check Your Knowledge

Knowledge Check Question

Question: Which of the following is the most likely cause of primary hypoaldosteronism?

Answer:

- **Addison disease. Correct.** Primary hypoaldosteronism is caused by disease in the adrenal gland itself, and the most common cause is primary adrenal deficiency (Addison disease), which also causes low cortisol levels.
- **Adrenal adenoma. Incorrect.** Aldosterone-secreting adrenal adenoma causes hyperaldosteronism, not hypoaldosteronism.
- **Diabetes mellitus. Incorrect.** Diabetes mellitus can cause secondary, not primary hypoaldosteronism by reducing renin secretion.
- **Nonsteroidal anti-inflammatory drugs. Incorrect.** Nonsteroidal anti-inflammatory drugs can cause secondary hyperaldosteronism by reducing renin secretion.
- **Type IV renal tubular acidosis. Incorrect.** Type IV renal tubular acidosis is a complication, not a cause of hypoaldosteronism.

Part 4 of 6: Congenital Adrenal Hyperplasia

Overview

Learning Objectives

This page covers the following learning objectives:

5. Describe the etiology and pathogenesis of primary adrenal insufficiency, including autoimmune destruction, infection, and genetic disorders.
6. Define Conn's syndrome (primary hyperaldosteronism) and describe the underlying pathophysiology, including aldosterone-producing adrenal adenomas and bilateral adrenal hyperplasia.

What Is Congenital Adrenal Hyperplasia?

Case Connection

At his 1-week well-child visit, JJ, a formula-fed infant, has lost about 12% of his birth weight. His exhausted parents report poor feeding, lethargy, and irritability. They have tried multiple bottles and nipples, and he just doesn't want to drink. He vomits a little bit, and he has had very few wet or dirty diapers. This is the couple's first baby. The pregnancy and delivery were uncomplicated. On exam, his temperature is normal and heart rate is 154 beats/min. His fontanelles are sunken, and his mucosal membranes are dry. You have not yet received the results of the newborn screening blood tests, so you ask the medical assistant to call the lab to check on them.

What test results are you concerned about? Consider your answer as you read, and we'll revisit JJ at the end of the page.

Congenital adrenal hyperplasia (CAH) is a group of rare autosomal recessive disorders caused by hereditary defects in the enzymes involved in adrenal steroid production, mainly cortisol. A low rate of cortisol synthesis leads to a high rate of adrenocorticotrophic hormone (ACTH) production, ultimately causing the characteristic adrenal enlargement (hyperplasia). These disorders are an important cause of pediatric hormone deficiencies and excesses and of congenital disorders of sexual organs. Because of their classic presentations and interesting endocrinology, they frequently appear on lists of differential diagnosis (and board exams).

Depending on the type of CAH, the disorder will not only cause lower cortisol levels, but also may alter production of other steroid hormones, such as aldosterone and androgenic

sex steroids. There are at least five types of CAH, but we'll focus on the three most common:

- 21-Hydroxylase deficiency
- 11 β -Hydroxylase deficiency
- 17 α -Hydroxylase deficiency

CAH has an overall prevalence of 1 in 16,000 births, but this rate is higher in some populations (eg, Ashkenazi Jews, some native Alaskan groups).

Normal Adrenal Hormone Production

The adrenal cortex is the production center for many important endogenous steroids. Recall that the three zones of the adrenal cortex produce three types of steroids:

- Most mineralocorticoids (such as aldosterone) are produced in the zona glomerulosa.
- Glucocorticoids like cortisol are produced in the zona fasciculata.
- Androgens like androstenedione and testosterone are made in the zona reticularis.

Figure 1 shows the three cortex zones, their histologies and regulation, and the hormone classes and hormones they produce.

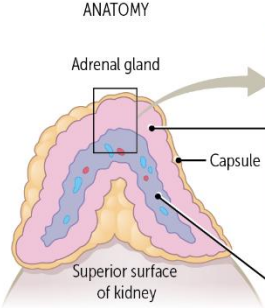
ANATOMY	HISTOLOGY		1 ^o REGULATION BY	HORMONE CLASS	1 ^o HORMONE PRODUCED
	CORTEX	Zona Glomerulosa	Angiotensin II	Mineralocorticoids	Aldosterone
		Zona Fasciculata	ACTH Corticotropin releasing hormone	Glucocorticoids	Cortisol
		Zona Reticularis	ACTH Corticotropin releasing hormone	Androgens	Dehydroepiandrosterone (DHEA)
	MEDULLA	Chromaffin cells	Preganglionic sympathetic fibers	Catecholamines	Epinephrine Norepinephrine

Figure 1. Image Credit: ScholarRx, used with permission.

Enzymes That Control Adrenal Steroid Synthesis

Let's analyze it: The zones of the adrenal gland are adjacent to one another, and the different cells interact to produce the final hormones. This complex network of hormone precursors and the controlling enzymes is shown in **Figure 2**.

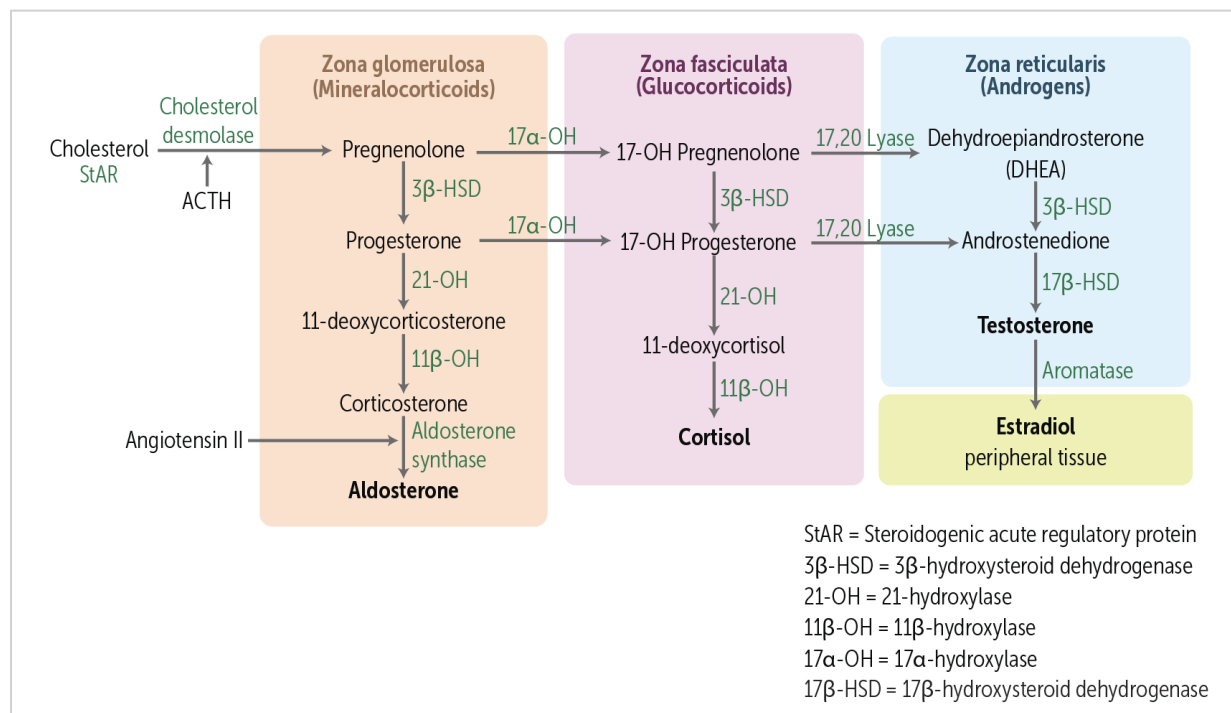


Figure 2. Image Credit: ScholarRx, used with permission.

- Each column indicates the hormone precursors and enzymes in their respective zona.
- The outside (superficial part) of the cortex is at the left (zona glomerulosa) and the inner (deep) part is on the right (zona reticularis).
- The final products are shown at the bottom: aldosterone, cortisol, and testosterone.
- Note that estradiol is not made in the adrenal gland but is converted from testosterone in the peripheral tissues.
- Also note that some of the intermediate precursors like 11-deoxycortisol and androstenedione are active hormones as well. They normally circulate at much lower levels than aldosterone, cortisol, estrogen, and the androgens. But in CAH, because of enzymatic blocks, the levels of these precursors can go up in the blood, causing hormonal effects.

Although this chart looks daunting, it is useful for understanding what happens when an enzyme is deficient. For example, you can see how a 21-hydroxylase deficiency would cause low levels of both aldosterone and cortisol. If you think about it, such a block in the pathway would also lead to a backup (excess) of the precursors above them on the chart



and shunt the reaction to the right side, increasing androgens. This is exactly what we see in a 21-hydroxylase deficiency.

Knowledge Check Question

Question: What are the three final products of adrenal steroid production?

Answer: Aldosterone, cortisol, and testosterone (along with some other androgens, such as androstenedione, which are both intermediate precursors and active hormones).

How Do Patients With Congenital Adrenal Hyperplasia Present?

Each type of CAH has a slightly different presentation because the enzyme affected is different. Also, because the gene mutations involved have varying degrees of penetrance, the range of symptoms experienced by a given patient may vary from mild disease presenting in adolescence to severe disease presenting at birth. But there are some common clinical patterns that can be seen. Surprisingly, the size of the adrenal glands does not usually lead to the diagnosis. Even though the adrenal glands are enlarged, it is most often the symptoms that lead to more investigation.

Clinical Features of Hormone Deficiencies

Failure to Thrive. Severe enzyme deficiencies often present in the first month of life with hormonal abnormalities that may cause failure to thrive and poor feeding. These particular abnormalities vary depending on which enzyme is deficient, but they are usually caused by a cortisol deficiency.

Hypoglycemia and Hypotension. All of the disorders cause **low cortisol levels**, which may present with hypoglycemia and hypotension.

Inability to Retain Salt and Water. Several of the disorders cause **low aldosterone levels**. This leads to renal salt wasting, dehydration, low blood volume, hypotension, and sometimes hyponatremia and hyperkalemia.

Clinical Features of Hormone Excess

As we know, when the body is unable to produce a particular hormone, sometimes it compensates by overproducing another hormone or precursor hormone.

- **Hyperpigmentation:** The low cortisol level causes an increase in anterior pituitary ACTH production. The high ACTH may cause hyperpigmentation. This is because of elevated ACTH and melanocyte-stimulating hormone, which is produced from the polypeptide precursor to ACTH, proopiomelanocortin.



- **Hypertension and hypokalemia:** On the other hand, some CAH disorders can increase hormonally active precursors to mineralocorticoids, such as 11-deoxycorticosterone. This can actually cause increased mineralocorticoid effects, including hypertension and hypokalemia.

Clinical Features of Virilization

Several of these disorders lead to abnormally high levels of androgens or estrogens. A high ratio of androgens to estrogens leads to virilization (also called masculinization).

- **Females:** In females, depending on the disorder, this can present at birth as ambiguous genitalia or clitoromegaly. There may be early growth of pubic hair or accelerated bone growth. Later in life, some forms give rise to irregular menses, infertility, and hirsutism.
- **Males:** These disorders are not usually detected in males at birth, because the genitalia appear normal. They may be diagnosed when boys undergo early signs of puberty, such as genital enlargement or pubic hair development.
- **Adolescents:** The clinical changes of virilization may cause growing children and adolescents to have significant issues regarding their gender identity and sexuality, so education and counseling are paramount.

Knowledge Check Question

Question: Which class of hormones most determines virilization?

Answer: Androgens (specifically dihydrotestosterone)



What Causes Congenital Adrenal Hyperplasia?

The CAH disorders are caused by genetic mutations inherited in an autosomal recessive pattern, and these mutations lead to deficiencies of enzymes that synthesize specific steroid hormones. Table 1 lists the exact genes associated with each of the enzyme deficiencies we are covering.

Type of CAH	Enzyme	Gene
21-Hydroxylase deficiency	21 α -Hydroxylase	CYP21A
11 β -Hydroxylase deficiency	11 β -Hydroxylase	CYP11B1
17 α -Hydroxylase deficiency	17 α -Hydroxylase/17,20 lyase	CYP17A1

The most common cause of CAH is a 21-hydroxylase deficiency, accounting for about 90% of cases. The next most common cause of CAH is an 11 β -hydroxylase deficiency, accounting for 5% to 8% of cases. And a 17 α -hydroxylase deficiency is a less common cause of CAH than the other two.

Knowledge Check Question

Question: What is the most common enzyme deficiency in CAH?

Answer: 21-Hydroxylase deficiency the most common enzyme deficiency in CAH.

How Do We Diagnose Congenital Adrenal Hyperplasia?

Now we'll look at how to diagnose CAH, focusing on differentiating the three main types. But before we do that, it's good to remember that all types typically show low cortisol levels. A common way of confirming this is with ACTH stimulation testing. This measures the level of cortisol in the patient's blood before and after administration of synthetic ACTH. If the adrenals are working properly, the cortisol level will rise. If the cortisol level does not adequately rise in response to ACTH administration, CAH is the most likely diagnosis. Now let's differentiate the different types of CAH.

21-Hydroxylase Deficiency

In CAH caused by 21-hydroxylase deficiency, insufficient production of cortisol and aldosterone occurs. Progesterone and 17-OH-progesterone levels then increase, because they are unable to develop into the final products of aldosterone and cortisol, respectively (**Figure 3**). This is the most common type of CAH.

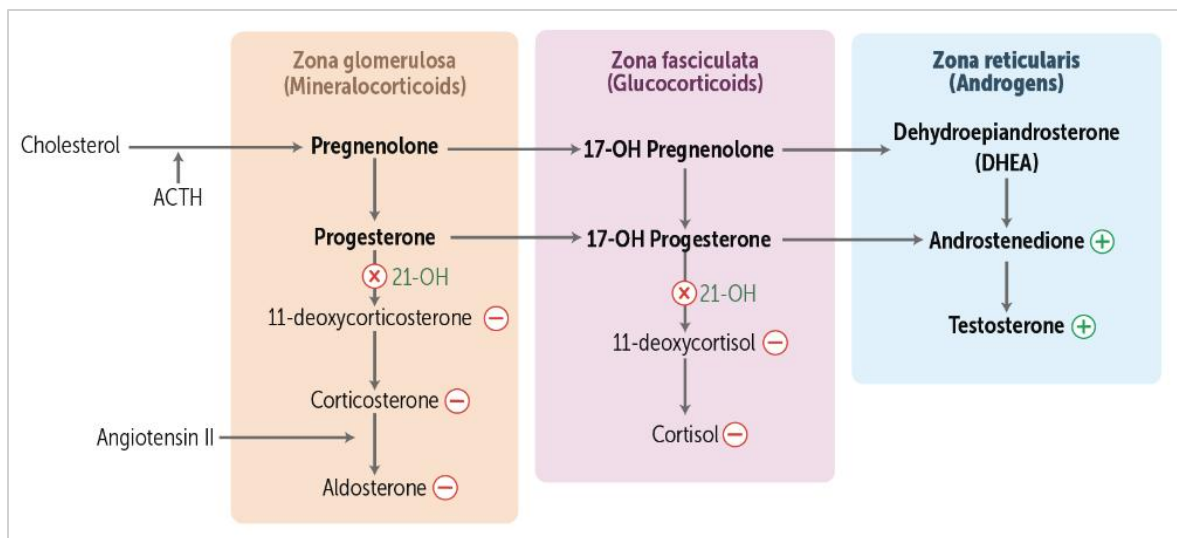


Figure 3. Image Credit: ScholarRx, used with permission.

Clinical Effects of Excessive Androgen Production. A hallmark of this disorder is excessive androgen production within the zona reticularis (eg, androstenedione, testosterone). This is because the buildup of precursor molecules is shunted to the pathways that produce sex hormones (see Figure 4).

How does it manifest? **In both sexes** there may be premature onset of puberty and short stature resulting from advanced bone age and premature closure of growth plates. The effects on the genitalia vary depending on the severity of the deficiency.

- **Males** are typically not affected by high levels of adrenal androgens because they are already exposed to high levels of testosterone and dihydrotestosterone.
- **Females**, however, are not typically exposed to such high levels of these androgens, which can cause virilization. This may lead to the male appearance of external genitalia, or an enlarged clitoris with or without partial fusion of labioscrotal folds; this can have the appearance of a male urethra. Adolescent females may have male-pattern hair growth on the face and body (hirsutism), polycystic ovarian syndrome, and abnormal frequency of menstrual periods. However, the internal female reproductive anatomy is not affected by this increased exposure to androgens.

Excessive androgen synthesis is the most common cause of genitals that are hard to distinguish as male or female—a condition known as genital ambiguity and seen in some patients with a 21-hydroxylase deficiency (Figure 4). The presence of testes or ovaries varies from patient to patient and helps identify sex.



Figure 4. Image Credit: ScholarRx, used with permission.

Knowledge Check Question

Question: What are some characteristic findings of virilization in a female infant?

Answer: Male appearance of external genitalia, abnormally large clitoris size, partial fusion of labioscrotal folds.

Clinical Effects of Low Aldosterone and Cortisol. Depending on the severity of the 21-hydroxylase deficiency, patients may also have renal salt wasting from the inability to produce aldosterone, an important stimulus to renal sodium reabsorption. This can cause dehydration and hypotension. Affected infants typically demand more feedings because they are dehydrated and have salt cravings. A combination of cortisol and aldosterone deficiencies may cause severe hyponatremia and hyperkalemia with vomiting, hypotension, and mental status changes, a condition termed adrenal crisis.

Clinical Correlation

Initial management for patients in adrenal crisis includes intravenous (IV) fluid resuscitation with sodium-containing fluids, electrolyte regulation, and cortisol and mineralocorticoid administration.

Newborn Screening. As a screen for 21-hydroxylase deficiency, 17-OH-progesterone is measured in all newborns in the United States.

11 β -Hydroxylase Deficiency

Looking at Figure 5, you might at first think that an 11 β -hydroxylase deficiency should look very much like a 21-hydroxylase deficiency, with lower levels of cortisol and aldosterone and higher levels of androgens.

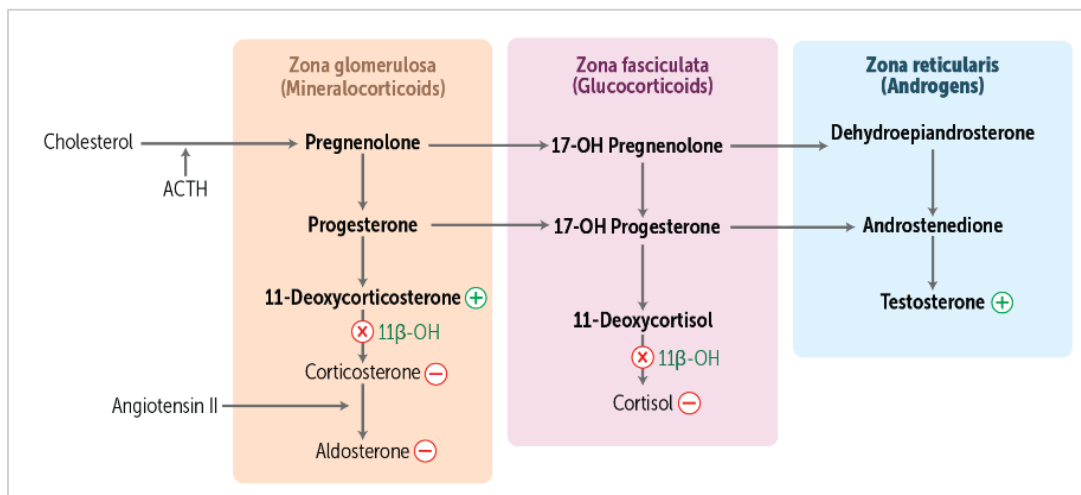


Figure 5. Image Credit: ScholarRx, used with permission.

But there is a big difference! The enzymatic block comes one step below **11-deoxycorticosterone**, so that hormone increases in this syndrome. Because the molecule has potent mineralocorticoid effects such as aldosterone, it causes high (not low) sodium reabsorption and high (not low) potassium excretion. So these patients instead get **hypertension and hypokalemia**, rather than the hypotension and salt wasting seen in 21-hydroxylase deficiency. Hypertension normally occurs in adulthood. Infants may present with adrenal crisis caused by low cortisol levels.

Patients with this syndrome have a similar range of virilizing and genitalia effects as those with a 21-hydroxylase deficiency.

17 α -Hydroxylase Deficiency

A deficiency of 17 α -hydroxylase is a less common cause of CAH than the previously mentioned deficiencies. The CYP17A1 gene is responsible for the activity of both 17 α -hydroxylase and 17,20 lyase. When CYP17A1 is mutated, there may be a partial or complete deficiency of either enzyme or both. The most common gene variant inhibits production of both 17 α -hydroxylase and 17,20 lyase (**Figure 6**).

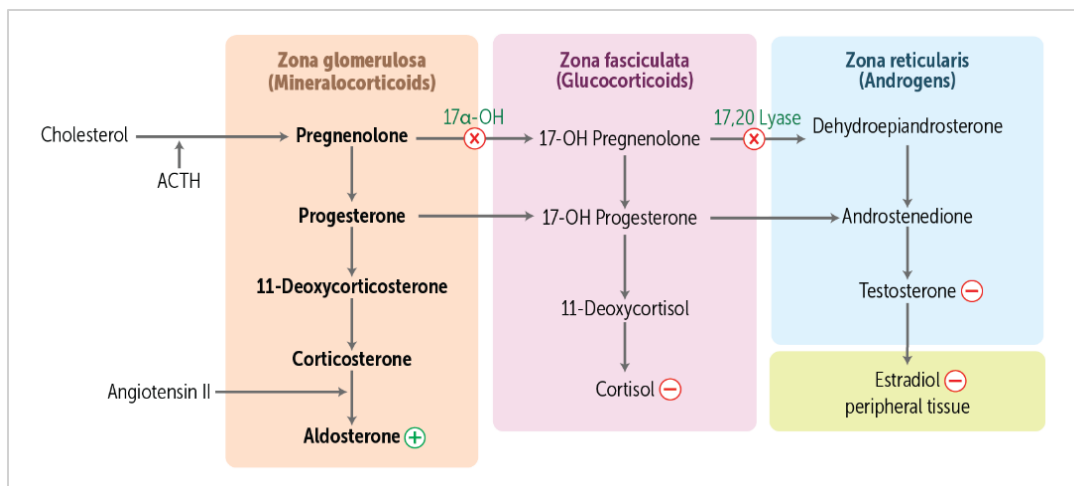


Figure 6. Image Credit: ScholarRx, used with permission.

Patients with this deficiency look a bit different. As you can see from **Figure 6**, the patients will have high mineralocorticoid levels but low androgen levels, unlike patients with the other syndromes we have discussed. At birth females will have normal genitalia, whereas males may have ambiguous genitalia. But at the normal time of puberty, females will not begin menstruating and usually will be infertile. They will also have less developed secondary sex characteristics (eg, breasts, pubic hair). Lower testosterone levels will cause males at puberty to have smaller penises and less pubic hair growth. They have less sperm development, so they often are infertile.

Wrapping it up, **Table 2** summarizes these three types of CAH as well as their associated hormonal abnormalities.

	Glucocorticoids	Mineralocorticoids/ blood pressure	Adrenal androgens	Presentation and lab tests
21-Hydroxylase	↓	Normal to ↓	↑	Female external genitalia ambiguous at birth Hyperandrogenism in adolescent females (hirsutism, irregular menses) Insufficient rise in cortisol on ACTH stimulation test Low aldosterone High progesterone and 17-OH-progesterone
11β-Hydroxylase	↓	↑	↑	Hypertension, hypokalemia Low cortisol and aldosterone on labs Increased androgens and 11-deoxycortisone
17α-Hydroxylase	↓	↑	↓	Low androgens Hypokalemia, hypertension Low androgens on labs High mineralocorticoids

Note that glucocorticoids are lower than normal in all three types of CAH. If you can remember that, then Table 3 may help you recall the other hormonal abnormalities, where an ↑ stands for each 1 in the enzyme name—you'll see that the arrow pattern corresponds to the up arrows for the corresponding hormones!



Deficiency	Mineralocorticoid/blood pressure	Adrenal androgens
21-Hydroxylase	Content	↑
11β-Hydroxylase	↑	↑
17α-Hydroxylase	↑	

How Do We Manage Congenital Adrenal Hyperplasia?

Because all patients with CAH are deficient in cortisol, glucocorticoid replacement is required. At birth, if presenting in adrenal crisis, patients need rapid treatment with high-dose glucocorticoids as well as IV saline. Once the patient is stable, then replace the deficient hormones to reduce the high adrenal androgen production (in 21-OH and 11β-OH deficiency).

- Patients with a **21-hydroxylase deficiency** may also require treatment with synthetic mineralocorticoids, such as fludrocortisone.
- Patients with an **11β-hydroxylase deficiency** are treated with cortisol replacement, as are patients with a 21-hydroxylase deficiency.
- For patient with a **17α-hydroxylase deficiency**, management includes cortisol replacement, hypertension control, and, for females, estrogen and progestins to treat the low androgen levels. Males receive testosterone to normalize the development of sexual characteristics.

Knowledge Check Question

Question: What is the best initial management for patients with 21-hydroxylase deficiency and symptoms of dehydration and hypotension?

Answer: Glucocorticoids with normal saline is used in all patients. Fludrocortisone is administered based on sodium and potassium levels.



Case Connection

Thinking back to JJ, what test results are you concerned about?

As you thought, baby JJ's newborn screen results are positive for CAH. You talk to the parents about the diagnosis and its treatment and briefly discuss the autosomal recessive heritability of the disease. You admit JJ so he can receive IV fluids and replacement steroids. Because you know that this condition has the potential to affect future children for the couple and others in their families, you arrange a consultation with the hospital genetics counselor.

Summary

- Congenital adrenal hyperplasia (CAH) is a condition resulting from genetic enzymatic deficiencies that decrease the production of adrenal cortical steroid hormones.
- CAH enzyme deficiencies cause low cortisol levels, leading to high adrenocorticotrophic hormone secretion and adrenal hyperplasia.
- The most common type of CAH is caused by a 21-hydroxylase deficiency resulting from gene mutations in CYP21A2.
- A 21-hydroxylase deficiency causes high levels of androgenic sex steroids, leading to virilization.
- An 11 β -hydroxylase deficiency causes elevated mineralocorticoid levels, leading to hypertension and hypokalemia; it also causes virilization.
- A 17 α -hydroxylase deficiency causes high mineralocorticoid levels but low androgen levels, leading to infertility and less development of secondary sexual characteristics at puberty.
- Patients with CAH require glucocorticoid replacement; they may also need mineralocorticoid, testosterone, or estrogen replacement.



Check Your Knowledge

Knowledge Check Question

Question: A previously healthy, term 10-day-old boy has presented to the emergency department for poor feeding. The patient has been a vigorous feeder, at times crying to be fed every hour. Over the past 2 days, he has become less active, and he started vomiting today. On exam, the patient has dry mucous membranes and darkening of the skin around his inguinal fold and in his diaper area. He has normal external genitalia with testes descended bilaterally. Serum studies show:

- Sodium: 129 mmol/L (reference range, 133-146 mmol/L)
- Potassium: 6.2 mmol/L (reference range, 3.7-5.9 mmol/L)
- Testosterone: 125 ng/mL (reference range, 20-75 ng/dL)
- Dehydroepiandrosterone sulfate: 450 µg/mL (reference range, 75-370 µg/dL)

Which of the following is the most likely mechanism of his high serum potassium level?

Answer:

- **Low adrenocorticotrophic hormone. Incorrect.** Adrenocorticotrophic hormone would be high because of the lack of cortisol production, so low adrenocorticotrophic hormone is incorrect.
- **Low aldosterone. Correct.** This patient has 21-hydroxylase deficiency and thus is unable to make aldosterone, which acts on the renal tubule to absorb sodium and water while excreting potassium into the urine. The deficiency of this enzyme has led to retention of potassium, as shown by the high serum value.
- **Low cortisol. Incorrect.** This may be present in 21-hydroxylase deficiency but do not directly affect potassium.
- **High dehydroepiandrosterone sulfate. Incorrect.** This may be present in 21-hydroxylase deficiency but do not directly affect potassium.
- **High testosterone. Incorrect.** This may be present in 21-hydroxylase deficiency but do not directly affect potassium.

Part 5 of 6: Pheochromocytoma

Overview

Learning Objectives

This page covers the following learning objectives:

7. Define pheochromocytoma and describe the pathophysiology of catecholamine hypersecretion.

What Is Pheochromocytoma?

Case Connection

NE is a previously healthy 67-year-old male who is in a motor vehicle accident and taken to the emergency department for evaluation. He is found to have a blood pressure (BP) of 156/100 mm Hg, which remains high even after his pain is controlled. On a CT scan done to look for skeletal trauma, a 1-cm tumor is seen in his right adrenal gland. He has not had any unusual symptoms. His BP last year was 120/78 mm Hg.

How would you approach your evaluation of NE's condition? Consider your answers as you read, and we'll revisit at the end of the page.

A pheochromocytoma is a rare neuroendocrine tumor most often found in the adrenal gland. This type of tumor is usually benign, but it can be malignant in rare cases.

Grossly, a pheochromocytoma is a dusky-appearing tumor, usually located in the adrenal gland (**Figure 1**).

Pheochromocytomas secrete **catecholamines** such as epinephrine and norepinephrine. These hormones promote the fight-or-flight response, so patients with this disorder may have symptoms associated with running from a bear (hypertensive, tachycardic, sweaty). The high catecholamines cause **secondary hypertension**, that is, blood pressure (BP) greater than 140/90 mm Hg caused by a specific, identifiable disease. It mostly affects adults and accounts for less than 0.5% of all cases of hypertension.

Why worry about it if it is so rare? There are two reasons: (1) its fight-or-flight symptoms make it easily confused with other conditions and (2) it is a favorite topic of attending physicians and exam writers, because it reflects normal physiology. Let's learn more about it.

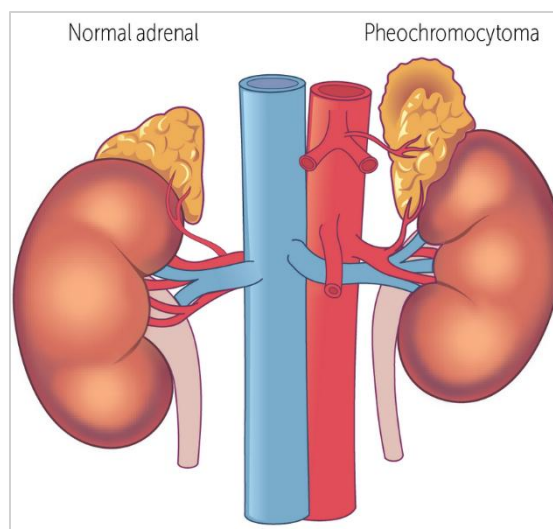


Figure 1. Image Credit: ScholarRx, used with permission.

Knowledge Check Question

Question: What hormones do pheochromocytomas secrete?

Answer: Pheochromocytomas secrete catecholamines, such as epinephrine and norepinephrine.

How Do Patients With Pheochromocytoma Present?

The presentation of pheochromocytoma can vary, ranging from relatively stable hypertension to hypertensive crisis or other critical events that can result in death. Given that variability, what do we look for?

Asymptomatic

About half of patients are asymptomatic. Sometimes the tumor is found inadvertently, such as on a CT scan done for another reason.

New or Worsening Hypertension

Many patients, however, present with new or worsening hypertension. Although most patients with high BP (>140/90 mm Hg) have essential (primary) hypertension, suspect secondary hypertension—such as in pheochromocytoma—when there is at least one of the following:

- There is a new elevation of BP at <25 years or >65 years.
- People present with very high BP (>180/120 mm Hg).



- Patients do not respond to conventional BP drugs (eg, require ≥ 3).
- Patients have a classic finding associated with one of the secondary causes.

Classic Triad

Half of patients will have at least some symptoms, but these can be vague (weakness, headaches). Only a small percentage have the triad of “classic symptoms,” which are episodic (come and go). These do appear on exams, so remember them even though most patients do not have them all:

- Headaches.
- Sweating.
- Palpitations secondary to tachycardia.

These classic symptoms can easily be confused with those of more common disorders, such as panic attacks or myocardial infarction, especially because activation of the sympathetic nervous system can induce panic attacks or a sense of impending doom in patients with pheochromocytoma. But remember that pheochromocytoma is a rare syndrome—it is more likely that your patient with hypertension and sweating just has essential hypertension with another cause of sweating (think horses vs zebras)!

Critical Incidents

These tumors often secrete catecholamines episodically, instead of in continuous, predictable amounts. This often results in paroxysmal symptoms. A patient with previously normal BP readings may suddenly develop a hypertensive crisis (>180 mm Hg systolic pressure or >120 mm Hg diastolic pressure). These unpredictable bursts of catecholamines can cause heart failure and cardiac arrhythmias, the latter of which can cause sudden death.

Knowledge Check Question

Question: Symptoms of pheochromocytoma: The six Ps

Answer: Symptoms of pheochromocytoma, the six Ps: Pressure (elevated BP), Pain (headaches), Palpitations (tachycardia), Perspiration (sweating), Pallor (paleness), and Panic attacks.

Knowledge Check Question

Question: What is the most common physical exam finding in pheochromocytoma?

Answer: Hypertension is the most common physical exam finding in pheochromocytoma.

What Is the Pathophysiology of Pheochromocytoma?

Knowledge Check Question

Question: Pheochromocytoma rule of 10s: 10% bilateral, 10% malignant, 10% calcification, 10% located outside the medulla (para-aortic, urinary bladder, thorax)

Answer: Pheochromocytoma rule of 10s

Genetics

About 30% to 40% of pheochromocytomas arise from an inherited gene mutation, and another 20% come from a somatic mutation that occurs after birth. Most of these mutations are random, but some happen as part of an inherited syndrome.

The most common distinct inherited syndromes are the **multiple endocrine neoplasia 2 (MEN2) syndrome** with RET gene mutations, neurofibromatosis type 1 with NF1 mutations, and the Von Hippel-Lindau disease with VHL mutations. Because of these, genetic testing is an important part of the evaluation of people with pheochromocytoma.

Clinical Correlation

MEN2 is characterized by pheochromocytoma, medullary thyroid cancer, and either parathyroid adenomas with hyperparathyroidism (MEN2A) or mucocutaneous neurofibromas, marfanoid habitus, and gastrointestinal symptoms (MEN2B).

Cellular Origins

These epithelial tumors arise from **neuroendocrine cells**, which are hormone-secreting cells with glandular and neural features. During normal fetal development, neural crest cells migrate to the adrenal medulla and sympathetic chain ganglia near the vertebral column. The adrenal **chromaffin cells** are derived from a neural crest and normally secrete catecholamines to support BP. Pheochromocytomas arise from these chromaffin cells.

Histology

This neoplasm is composed largely of polygonal/spindle cells arranged in nests within a rich vascular network (**Figure 2**). The nuclei of the cells are oval to round and frequently contain prominent nucleoli. The cytoplasm is red-purple.

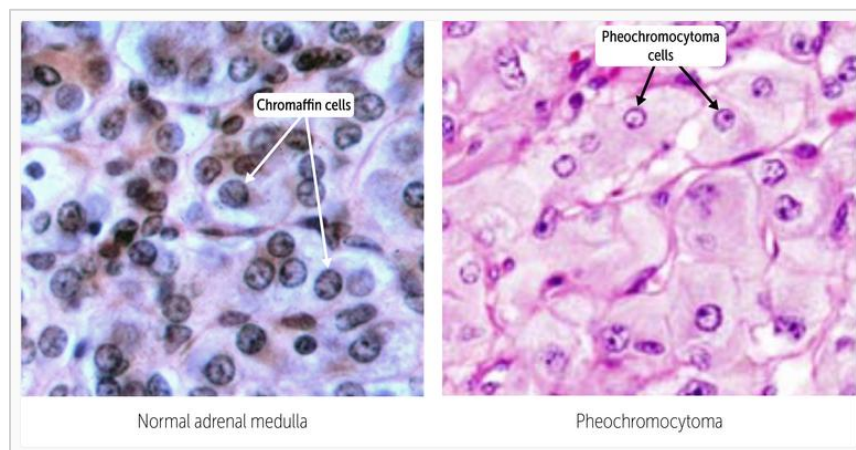


Figure 2. Image Credit: ScholarRx, used with permission.

Knowledge Check Question

Question: What type of cells do pheochromocytomas arise from?

Answer: Pheochromocytomas arise from chromaffin cells of the adrenal medulla.

Physiology

What causes the classic symptoms seen in some patients? Increased activation of the sympathetic nervous system causes these classic symptoms:

- The tachycardia is caused by β_1 -adrenergic stimulation.
- α_1 -Adrenergic-mediated vasoconstriction can cause patients to appear pale (**pallor**), as blood is shunted away from the skin.
- The **sweating** is mediated by central activation of heat loss mechanisms by the catecholamines; it is also in response to the increased thermogenesis and vasoconstriction.

Complications

The complications associated with pheochromocytoma are mainly caused by severe hypertension, which may damage the brain, heart, and kidneys. Hypertensive crisis can lead to stroke, myocardial infarction, and heart failure. Cardiac arrhythmias are another



severe complication, with up to one-third of patients experiencing sudden death before diagnosis. Surgical removal of pheochromocytomas prevents these complications.

Clinical Correlation

In September 1955, the US President Dwight D. Eisenhower had an acute myocardial infarction; he would later die in 1969 of ischemic cardiomyopathy. When the autopsy revealed a 1.5-cm pheochromocytoma in his left adrenal gland, investigators realized that the hypertension and systolic and diastolic BP spikes that Eisenhower had experienced throughout his life may have resulted from his tumor.

How Do We Diagnose Pheochromocytoma?

To definitively diagnose pheochromocytoma, we need biochemical evidence of an excessive release of catecholamines and anatomical localization of the tumor. So, after documenting the high BP, we conduct lab tests for high catecholamines, followed by radiologic studies.

Laboratory Tests

The best initial tests are for **metanephrines**, **vanillylmandelic acid (VMA)**, and **homovanillic acid (HVA)** in 24-hour urine collections. These are urine metabolic products that are generated as catecholamines are broken down. When a pheochromocytoma produces high levels of catecholamines, these metabolites are excreted in the urine. However, because of the paroxysmal secretion of catecholamines, patients must collect their urine for a full 24 hours to obtain a reliable test sample.

Elevated plasma levels of metanephrines and catecholamines support the diagnosis.

If urine and blood tests are indeterminate, the next step in testing includes a clonidine suppression test. Clonidine is a sympathetic blocking medication that typically decreases catecholamine release in a normal adrenal medulla. However, a pheochromocytoma generally operates independently and does not rely on input from the sympathetic nervous system. So, even after the administration of clonidine, catecholamines remain elevated in patients with pheochromocytoma.

Knowledge Check Question

Question: Elevations of which substances in a 24-hour urine test are most diagnostic for a patient with pheochromocytoma?

Answer: Levels of metanephrines, VMA, and HVA are elevated in the urine.

Radiology Studies

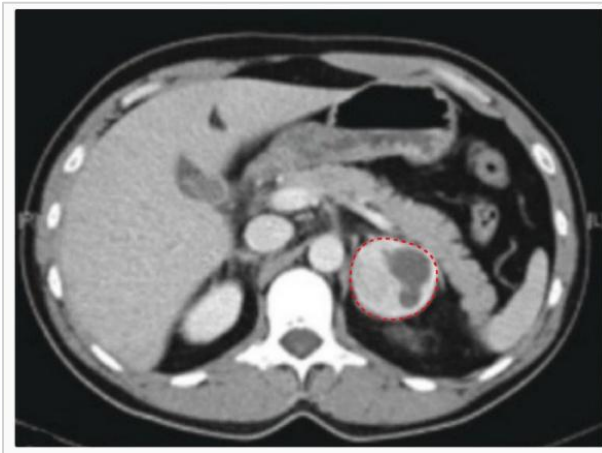


Figure 3. Image Credit: ScholarRx, used with permission.

Once the presence of the tumor is confirmed biochemically, we need to find where the tumor is. This is most often done by CT scanning of the abdomen (**Figure 3**).

In **Figure 3**, you can see a 5-cm pheochromocytoma (red dashed outline) in the left adrenal gland. The tumors may be quite small, so if these scans do not reveal the tumor, then a radiolabeled iodine–meta-iodobenzylguanidine (MIBG) scan is done (**Figure 4**).

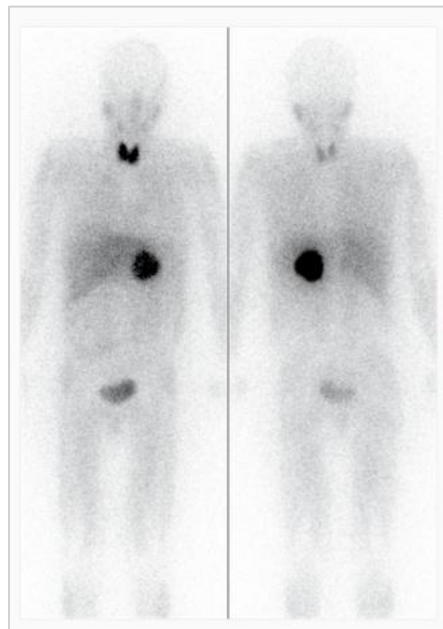


Figure 4. Image Credit: ScholarRx, used with permission.

The isotope is taken up by the tumor and seen on the nuclear medicine scanner. Note the large pheochromocytoma on the patient's left adrenal gland. The uptake of the iodine-123 (I-123) in the thyroid is an artifact caused by dissociation of the I-123 from the MIBG.



Differential Diagnosis

Pheochromocytoma is an uncommon cause of secondary hypertension (hypertension caused by a specific disorder). Recall that secondary hypertension accounts for only 5% of patients with hypertension—the rest have essential (primary) hypertension. Table 1 lists some other common causes of secondary hypertension, from most to least common.

Cause	Findings	Etiology
Chronic kidney disease	Edema, elevated serum creatinine	Glomerular or interstitial disease
Renal artery stenosis	Abdominal or flank bruit, hypokalemia	Atherosclerosis, fibromuscular dysplasia
Hyperaldosteronism	Hypokalemia	Adrenal adenoma, adrenal hyperplasia
Cushing syndrome	Buffalo hump, moon facies, striae	Prescribed corticosteroid drugs, adrenal or pituitary tumor
Pheochromocytoma	Tachycardia, sweating, palpitations, headache	Chromaffin cell neoplasia

How Do We Manage Pheochromocytoma?

Most adrenal pheochromocytomas can be **resected surgically** by using laparoscopic removal. Because the resection procedure can cause the tumor to release large amounts of catecholamines, the patient should first receive α -adrenergic blockade with phenoxybenzamine, followed by β -blockade with a drug such as metoprolol to block the action of the large amounts of catecholamines at the receptor level.

Clinical Correlation

It's important to do the α -blockade before the β -blockade. Use of a β -blocker without preexisting α -blockade can block the vasodilating β_2 -adrenergic receptors, leaving unopposed α -adrenergic vasoconstriction. This can lead to dangerously high BP.

What is the **prognosis** for patients with pheochromocytoma? Because 10% of pheochromocytomas are malignant, prompt diagnosis is essential, as is removal of the tumor. Five-year survival is 95% for patients with benign tumors and 50% for patients with malignant tumors.

Knowledge Check Question

Question: What medical treatment must be administered before resection of pheochromocytoma?

Answer: α -Adrenergic blockade (phenoxybenzamine), followed by β -blockade (eg, metoprolol), must be administered before resection of pheochromocytoma.

Case Connection

Thinking back to NE, how would you approach the evaluation of his condition? Finding an adrenal medulla tumor along with new hypertension made the clinicians think about the rare tumor pheochromocytoma. After NE recovered from the accident, his urine was tested; the results showed elevated levels of metanephrines. NE was treated first with phenoxybenzamine to reduce catecholamine-induced vasoconstriction, then with metoprolol to reduce the heart rate. The tumor was resected with an endoscope, and his BP immediately came back down to normal. Because the tumor was shown to be benign, his prognosis is excellent.

Summary

- Pheochromocytomas are rare, usually benign, tumors of the adrenal medulla.
- Pheochromocytomas secrete catecholamines that cause hypertension.
- Pheochromocytoma is usually diagnosed in adults with severe, worsening hypertension.
- Fifty percent of patients are asymptomatic.
- In symptomatic patients there may be paroxysmal symptoms of catecholamine release, such as palpitations, excess sweating, pallor, and panic attacks.
- Approximately 30% are associated with familial genetic syndromes such as multiple endocrine neoplasia 2.
- Pheochromocytoma is diagnosed by high levels of urine or plasma catecholamines or their metabolites, then localized by CT or meta-iodobenzylguanidine scanning.
- Pheochromocytoma is treated with surgical resection; pretreatment with α -blockers, followed by β -blockers is required before surgery.
- Complications of pheochromocytoma are often caused by hypertensive damage to the brain, heart, or kidneys; sudden death from arrhythmia can occur.



Check Your Knowledge

Knowledge Check Question

Question: A 50-year-old female presents to clinic reporting multiple episodes of headaches and palpitations. She recently visited the emergency department because of a panic attack with profuse sweating. Vital signs show elevated blood pressure at 180/120 mm Hg. A 24-hour urine collection demonstrates elevated urine metanephrines. Which of the following is the most likely tumor origin?

Answer:

- **Allantois. Incorrect.** The allantois gives rise to the urachus.
- **Apical ectodermal ridge. Incorrect.** The apical ectodermal ridge is ectoderm at the end of developing limbs.
- **Bulbus cordis. Incorrect.** The bulbus cordis develops into the smooth outflow tract of the left and right heart ventricles.
- **Neural crest cells. Correct.** Neural crest cells that are present within the adrenal medulla and in tissues along the paravertebral axis develop into chromaffin cells of the adrenal medulla that give rise to pheochromocytomas.
- **Notochord. Incorrect.** The notochord later becomes the nucleus pulposus.



Part 6 of 6: Neuroblastoma

Overview

Learning Objectives

This page covers the following learning objectives:

2. Describe the pathophysiology of endogenous Cushing's syndrome, including adrenal adenomas, pituitary adenomas, and ectopic ACTH production.
3. Describe the pathology of adrenal adenomas and carcinomas.

What is Neuroblastoma?

Case Connection

Your cousin asks if you will watch her toddler, TJ, while she goes to a conference. "I'd ask Mom to watch him," she says. "But he hasn't really been eating well lately, and I know she would just let him eat marshmallows all weekend." While babysitting, you notice a mass in TJ's belly. He seems fussy, too, so you make note to mention it to your cousin when she returns. "He didn't really want to eat much," you tell her. "And it feels like there's a lump in his belly." You show her what you noticed, and she calls their clinician to make an appointment.

How would you approach your evaluation of TJ's condition? Consider your answer as you read, and we'll revisit TJ at the end of the page.

Neuroblastoma is an uncommon childhood neuroendocrine cancer that is most often found in the adrenal medulla, or sometimes in the abdomen. It secretes catecholamines, the circulating neurotransmitters that are responsible for such stress responses as a pounding heart, muscle tensing, and sweating—in other words, fight-or-flight responses.

Most cases of neuroblastoma occur in young children, in whom it is the third most common cancer (after lymphoma and brain cancers). Forty percent are diagnosed before they are 1 year old. The incidence is about 10 cases per million (vs 50 per million for acute lymphocytic lymphoma, the most common pediatric malignancy).

Knowledge Check Question

Question: What neurotransmitters does a neuroblastoma secrete?

Answer: A neuroblastoma secretes catecholamines.



How Do Patients With Neuroblastoma Present?

Most neuroblastomas are seen in children younger than 4 years. Presentation can vary depending on location and size of the neuroblastoma as well as extent of spread. The child often won't experience any pain until the cancer has metastasized or the tumor is large enough to apply pressure on surrounding tissues.

The child may present with abdominal distention and a palpable, irregular abdominal mass that tends to cross the midline (Figure 1). Commonly, the child's parent notices the abdominal mass while bathing the child.

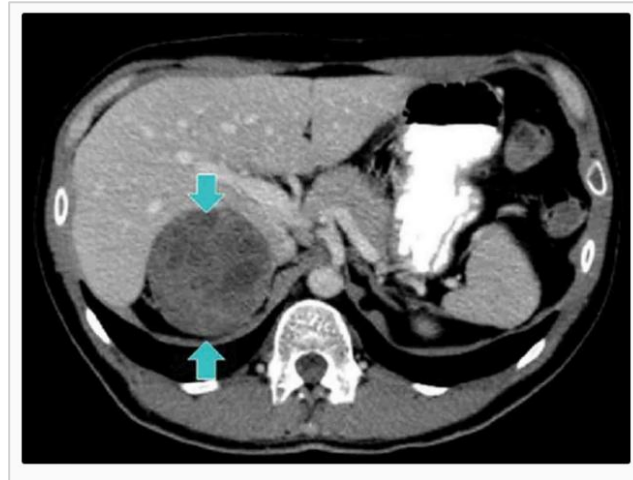


Figure 1. Image Credit: ScholarRx, used with permission.

Knowledge Check Question

Question: By what age do most patients present with neuroblastoma?

Answer: Most patients present with neuroblastoma by age 4 years.

What Is the Pathophysiology of Neuroblastoma?

Neuroblastoma usually is caused by a genetic mutation, though we don't know exactly what triggers it. What we do know is that it is an embryologic malignancy associated with cancerous mutations in neural crest cells. Now, how does this happen?

Genetic Features

In most patients with neuroblastoma, the MYCN oncogene is amplified and overexpressed. Oncogenes promote uncontrolled cell division and ultimately inhibit programmed cell death (apoptosis). In other words, cells uncontrollably divide, continue to grow, and develop further



mutations, leading to cancer. *MYCN* amplification and overexpression are associated with more aggressive and advanced forms of the disease; patient survival is much lower in these cases.

Genetic diseases associated with neuroblastoma include Hirschsprung disease (nerves missing from portions of the intestine, associated *RET* mutation) as well as neurofibromatosis type 1 (neurofibromas, Lisch nodules in the pigment of the eye, associated mutation of *NF1* on chromosome 17).

Cellular and Anatomic Origins

During normal fetal development, **neural crest cells** migrate to the adrenal medulla and sympathetic chain ganglia near the vertebral column. Neuroblastoma arises from neural crest cells that become cancerous. The tumor can develop anywhere along the sympathetic chain, but most often is found in the **adrenal medulla**.

However, it can also develop elsewhere in the abdomen. This has to do with the migration pattern of neural crest cells. While many neural crest cells migrate to the adrenal gland to give rise to the chromaffin cells of the adrenal medulla, neural crest cells also migrate to the abdomen in the embryo to form other structures, such as sympathetic ganglia and the submucosal plexus of the gut.

Knowledge Check Question

Question: What anatomic location is most common for neuroblastoma to develop?

Answer: The adrenal medulla is the most common anatomic location where these tumors develop.

Histology

Hematoxylin and eosin (H&E) staining of the tissue shows numerous small, round, blue cells, which form circular patterns surrounding neurofibrils; these are termed Homer-Wright rosettes (Figure 2).

Immunohistochemistry demonstrates expression of bombesin, a tumor marker seen in only a few cancers, and a reaction to neuron-specific enolase (NSE), an antibody that distinguishes neural tissue.

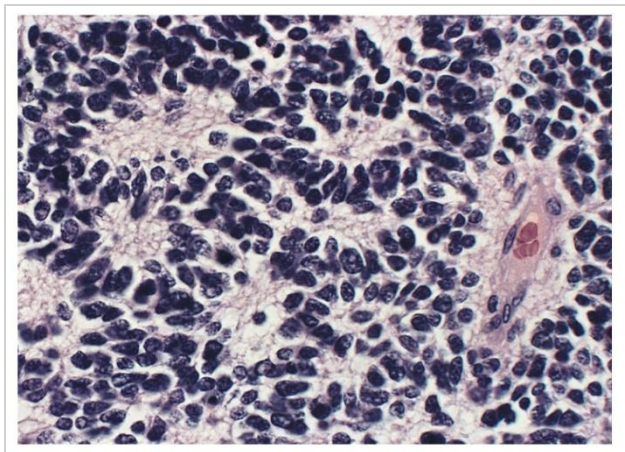


Figure 2. Image Credit: ScholarRx, used with permission.

Knowledge Check Question

Question: What is the classic histologic finding of neuroblastoma?

Answer: H&E staining of the tissue shows Homer-Wright rosettes—numerous small, round, blue cells that form circular patterns surrounding neurofibrils.

Complications

The potential complications from neuroblastoma include:

- **Metastasis:** More than 50% of patients have metastatic disease at presentation. The cancer can metastasize to the liver, bone marrow, bone, and skin.
- **Hypertension:** Neuroblastomas grow rapidly and may cause hypertension, either from renal artery compression or from catecholamines released from the tumor.
- **Constipation:** Constipation can occur if the tumor compresses the bowel. Neuroblastoma in the chest may present with wheezing, whereas metastasis to the bones may present with aching.
- **Opsoclonus-myoclonus syndrome:** Children with neuroblastoma can develop opsoclonus-myoclonus syndrome (“dancing eyes, dancing feet”). A concerned parent may explain that the child’s eyes have been moving fast and the child’s muscles have been twitching. Opsoclonus-myoclonus syndrome causes rapid, involuntary eye movements and muscle twitching. This is a rare paraneoplastic syndrome associated with neuroblastoma.
- **Spinal cord compression:** Rarely, neuroblastomas that arise in the paraspinal ganglia may compress the spinal cord and cause paralysis. That said, neuroblastomas in the paraspinal ganglia are quite rare. Most people with paragangliomas are asymptomatic; among people who do have symptoms, those symptoms vary and depend on the location of the tumor.



How Do We Diagnose Neuroblastoma?

Beyond the physical exam, there are a variety of tests that can help diagnose neuroblastoma.

Laboratory Studies

Most neuroblastomas secrete catecholamines, so serum dopamine, homovanillic acid (HVA), or vanillylmandelic acid (VMA) levels may be elevated.

Urine testing may show elevated urinary catecholamine metabolites, including HVA and VMA (metabolites of epinephrine and norepinephrine).

Imaging

The best initial test is abdominal ultrasound, which involves no radiation, and will determine whether further imaging is indicated. Once the tumor is detected, further staging is done using CT or MRI, as the tumor may be widespread.

Also helpful in diagnosis is the meta-iodobenzylguanidine (MIBG) scan, which involves uptake of the radioactively labeled chemical by the tumor. Uptake is positive in about 90% of tumors.

Biopsy

Tissue biopsy of the tumor is required for diagnosis.

Differential Diagnosis

Other pediatric tumors presenting similarly to neuroblastoma with an abdominal mass may include Wilms tumor, lymphomas, hepatoblastoma, rhabdomyosarcoma, and renal cell carcinoma. However, most of these tumors are smooth and do not cross the midline.

Knowledge Check Question

Question: What are the two urinary catecholamine metabolites most often measured in testing for neuroblastoma?

Answer: Homovanillic acid (HVA) and vanillylmandelic acid (VMA) are the two urinary catecholamine metabolites most often measured in testing for neuroblastoma.

How Do We Manage Neuroblastoma?

Treatment options depend on the patient's age, stage of disease, location of the tumor, and whether there is an abnormal genetic component to the tumor.

For example, for a child with a **low-risk tumor** without metastasis and MYCN amplification, surgery may be the only therapy needed to completely remove the tumor. If the tumor cannot be completely resected, chemotherapy and radiation therapy may be needed.



Children with **high-risk tumors** (usually stage III or IV, or stage II with MYCN amplification), on the other hand, will likely need a stem cell transplant and immunotherapy.

Clinical Correlation

Platinum-based chemotherapy treatments, such as cisplatin or carboplatin, can cause significant hearing loss in children with neuroblastoma. However, these agents greatly decrease mortality in these patients, so they are often still used. Hearing tests may be periodically done during chemotherapy treatment with these agents.

What is the prognosis? Generally, the outlook for patients with neuroblastoma is poor. Up to 50% of patients with high-risk disease do not respond to treatment, and unfortunately, disease recurrence is common.

Knowledge Check Question

Question: If the tumor cannot be resected, what additional treatments may be needed?

Answer: Chemotherapy and radiation.

Case Connection

Thinking back to TJ, how would you approach the evaluation of his condition?

Several types of abdominal tumors occur in children, but neuroblastoma is most common. Imaging such as an ultrasound would be a useful first step to determine the possible origin and extent of the tumor. A complete blood count and urinalysis are also helpful. Your cousin calls after the testing is done and tells you that TJ is scheduled for a biopsy later this week. The clinician thinks TJ may have a neuroblastoma. You offer to help your cousin find reliable information about neuroblastoma and make plans to meet for lunch tomorrow to discuss.

Summary

- Neuroblastoma is a rare, catecholamine-secreting neuroendocrine tumor that arises from neural crest cells and most often occurs in the adrenal gland.
- Neuroblastoma is usually diagnosed in children presenting with an irregular abdominal mass that may cross the midline.
- Neuroblastomas with MYCN oncogene amplification and overexpression are typically more aggressive.



- Neuroblastoma histology shows numerous small, round, blue cells forming circular patterns surrounding neurofibrils (Homer-Wright rosettes).
- Neuroblastoma can be diagnosed with high serum catecholamines, meta-iodobenzylguanidine scans, or biopsy, which is definitive.
- Neuroblastoma is treated with a combination of surgical resection, chemotherapy, radiation, stem cell transplantation, or immunotherapy.
- Many patients with high-risk neuroblastoma do not respond to treatment, and disease recurrence is common.

Check Your Knowledge

Knowledge Check Question

Question: A 2-year-old male presents to the pediatrician after his mother noticed an irregular abdominal mass crossing the midline. On exam, the physician notes the patient's dancing eye movements and unpredictable leg twitching. What are the biopsy results most likely to show?

Answer:

- **Circular, basophilic psammoma bodies. Incorrect.** Circular, basophilic psammoma bodies are collections of calcium associated with mesothelioma, meningioma, papillary carcinoma of the thyroid, and serous papillary cystadenocarcinoma of the ovary.
- **Clusters of azurophilic granules forming elongated needles. Incorrect.** Clusters of azurophilic granules forming elongated needles are also known as Auer rods and are found in acute myeloid leukemia.
- **Mucin-filled cells with nuclei on the periphery forming a signet-ring appearance. Incorrect.** Mucin-filled cells with nuclei on the periphery forming a signet-ring appearance are found in gastric carcinoma.
- **Sheets of lymphocytes and tangible body macrophages. Incorrect.** Sheets of lymphocytes and tangible body macrophages describes the “starry-sky” appearance of Burkitt lymphoma.
- **Small, round, blue cells forming circular patterns surrounding neurofibrils. Correct.** This description fits the classic histologic finding of Homer-Wright rosettes in neuroblastoma.